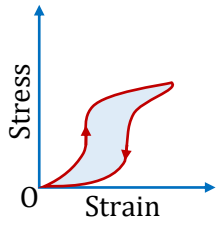
EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Elasticity

1. The diagram shows a stress-strain graph for a rubber band.



Consider of the following statements.

I. It will be easier to compress this rubber than expand it.

II. Rubber does not return to its original length after it is stretched.

III. The rubber band will get heated if it is stretched and released.

Which of these can be deduced from the graph

(A) III only (B) II and III (C) I and III (D) I only

PEH159

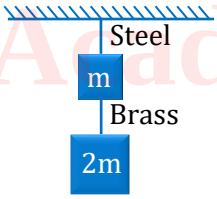
2. Overall changes in volume and radii of a uniform cylindrical steel wire are 0.2% and 0.002% respectively when subjected to some suitable force. Longitudinal tensile stress acting on the wire is:- ($Y = 2.0 \times 10^{11}\ Nm^{-2}$)

(A) $3.2 \times 10^{9}\ Nm^{-2}$ (B) $3.2 \times 10^{7}\ Nm^{-2}$ (C) $3.6 \times 10^{9}\ Nm^{-2}$ (D) $3.9 \times 10^{8}\ Nm^{-2}$

PEH081

3. If the ratio of lengths, radii and Young's moduli of steel and brass wires in the figure are a, b and c respectively, then the corresponding ratio of increase in their lengths is :-

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(A) $2\ a^2c/b$ (B) $3\ a/2b^2c$ (C) $2\ ac/b^2$ (D) $3\ c/2ab^2$

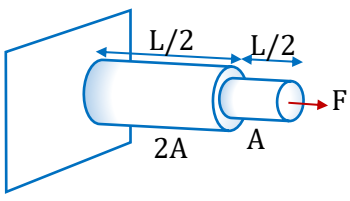
PEH082

4. A 100 m long wire having cross-sectional area $6.25 \times 10^{-4}\ m^2$ and Young's modulus is $10^{10}\ Nm^{-2}$ is subjected to a load of 250 N, then the elongation in the wire will be :

(A) $6.25 \times 10^{-3}\ m$ (B) $4 \times 10^{-4}\ m$ (C) $6.25 \times 10^{-6}\ m$ (D) $4 \times 10^{-3}\ m$

PEH175

5. The bar shown in the figure is made of a single piece of material. It is fixed at one end and consists of two segments of equal length $\frac{L}{2}$ but different cross-sectional area A and $2A$.



A = Area of cross section

What is the change in length of the entire system under the action of an axial force F ?

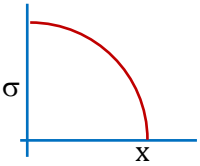
[Consider the shape of joint to remain circular, Y is young's modulus].

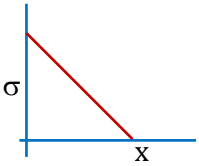
(A) $\frac{3FL}{4AY}$ (B) $\frac{3FL}{8AY}$ (C) $\frac{3FL}{2AY}$ (D) $\frac{2FL}{3AY}$

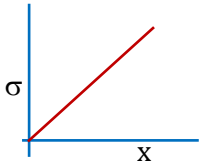
PEH083

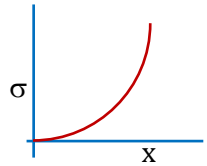
For More Join: @IITJEE_Advanced

6. A uniform rod rotating in gravity free region with certain constant angular velocity. The variation of tensile stress with distance x from axis of rotation is best represented by which of the following graphs.

(A) 

(B) 

(C) 

(D) 

PEH085

7. A solid sphere of radius R made of material of bulk modulus K is surrounded by a liquid in a cylindrical container. A massless piston of area A floats on the surface of the liquid. When a mass m is placed on the piston to compress the liquid, the fractional change in the radius of the sphere $\delta R/R$ is:-

(A) mg/AK (B) $mg/3AK$ (C) mg/A (D) $mg/3AR$

PEH087

8. When a steel wire fixed at one end is pulled by a constant force F at its other end, its length in equilibrium increases by l. Which of the following statements is not correct?

(A) Work done by the external force is Fl.

(B) Some heat is produced in the wire in the process.

(C) The elastic potential energy of the wire is $Fl/2$.

(D) The heat produced is equal to half of the elastic potential energy of the wire.

PEH090

9. An aluminium rod with Young's modulus $Y = 7.0 \times 10^{10}\ N/m^2$ undergoes elastic strain of 0.04%. The energy per unit volume stored in the rod in SI unit is:

(A) 5600 (B) 8400 (C) 2800 (D) 11200

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PEH177

10. A cylindrical wire of radius 1 mm, length 1 m, Young's modulus = $2 \times 10^{11}\ N/m^2$, Poisson's ratio $\mu = \pi/10$ is stretched by a force of 100 N. Its radius will become

(A) 0.99998 mm (B) 0.99999 mm (C) 0.99997 mm (D) 0.99995 mm

PEH091

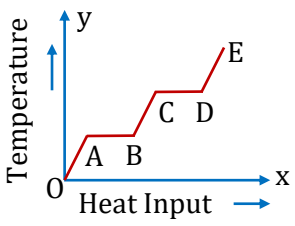
11. The length of wire becomes l_1 and l_2 when $100N$ and 120 N tensions are applied respectively. If $10l_2 = 11l_1$, the natural length of wire will be $\frac{1}{x} l_1$. Here the value of x is ___.

(A) 2 (B) 5 (C) 8 (D) 10

PEH178

Calorimetry

12. A solid material is supplied with heat at a constant rate. The temperature of material is changing with heat input as shown in the figure. What does slope DE represents?



(A) latent heat of liquid

(B) latent heat of vapour

(C) heat capacity of vapour

(D) inverse of heat capacity of vapour

PEH058

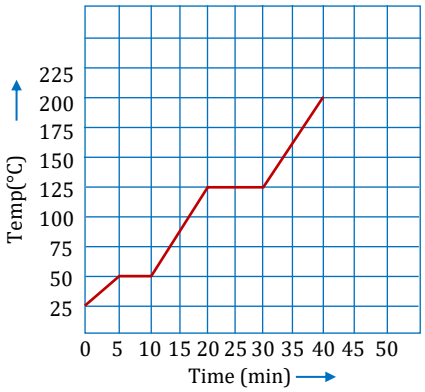
For More Join: @IITJEE_Advanced

13. In a container of negligible heat capacity 100 gm of a liquid at 20°C is heated. Specific heat of the liquid varies with temperature given as $s = (100T + 500)\ J/kg\text{-}°C$, where T is in °C. Find the amount of heat required to raise the temperature of the liquid to 40°C.

(A) 5000 J (B) 6000 J (C) 7000 J (D) 8000 J

PEH060

14. The graph shown in the figure represent change in the temperature of 5 kg of a substance as it absorbs heat at a constant rate of 42 $kJ\ min^{-1}$. The latent heat of vaporization of the substance is :



(A) 630 $kJ\ kg^{-1}$ (B) 126 $kJ\ kg^{-1}$ (C) 84 $kJ\ kg^{-1}$ (D) 12.6 $kJ\ kg^{-1}$

PEH061

15. Two bodies of specific heats S_1 and S_2 having same heat capacities are combined to form a single composite body. Specific heat of composite body is -

(A) $S_1 + S_2$ (B) $\frac{S_1+S_2}{2}$ (C) $\frac{2S_1S_2}{S_1+S_2}$ (D) $\frac{1}{S_1}+\frac{1}{S_2}$

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PEH062

16. A gas heater raises temperature of 1kg of water by 30°C in 10min. How long will it take the same heater to raise 3kg of oil through the same temperature rise if the specific heat of water is twice that of oil (assuming that there is no heat loss)

(A) 15 min (B) 5 min (C) 30 min (D) 60 min

PEH064

17. A continuous flow water heater (geyser) has an electrical power rating = 2 kW and efficiency of conversion of electrical power into heat = 80%. If water is flowing through the device at the rate of 100 cc/sec, and the inlet temperature is 10°C, the outlet temperature will be

(A) 12.2°C (B) 13.8°C (C) 20°C (D) 16.5°C

PEH069

18. Find the amount of heat supplied to decrease the volume of an ice water mixture by 1 cm^3 without any change in temperature. ($\rho_{ice} = 0.9\ \rho_{water}$, $L_{ice} = 80\ cal/gm$).

(A) 360 cal (B) 500 cal (C) 720 cal (D) none of these

PEH074

19. A thermally insulated vessel contains some water at 0°C. The vessel is connected to a vacuum pump to pump out water vapour. This results in some water getting frozen. It is given latent heat of vaporization of water at 0°C = $21 \times 10^5\ J/kg$ and latent heat of freezing of water = $3.36 \times 10^5\ J/kg$. The maximum percentage amount of water that will be solidified in this manner will be:-

(A) 86.2% (B) 33.6% (C) 21% (D) 24.36%

PEH076

For More Join: @IITJEE_Advanced

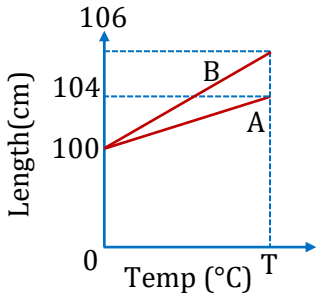
20. A lead ball at 30°C is dropped from a height of 6.2 km. The ball is heated due to the air resistance and it completely melts just before reaching the ground. The molten substance falls slowly on the ground. Calculate the latent heat of fusion of lead. Specific heat capacity of lead = 126 $J\,kg^{-1}°C^{-1}$ and melting point of lead = 330°C. Assume that any mechanical energy lost is used to heat the ball. Use g = 10 $m\,s^{-2}$.

(A) $6 \times 10^3\ J\,kg^{-1}$ (B) $1.2 \times 10^4\ J\,kg^{-1}$ (C) $1.8 \times 10^4\ J\,kg^{-1}$ (D) $2.4 \times 10^4\ J\,kg^{-1}$

PEH161

Thermal Expansion

21. The variation of lengths of two metal rods A and B with change in temperature are shown in figure. The ratio of $\frac{\alpha_A}{\alpha_B}$ is



(A) $\frac{3}{2}$ (B) $\frac{2}{3}$

(C) $\frac{4}{3}$ (D) $\frac{3}{4}$

PEH101

22. Two rods one of aluminium of length l_1 having coefficient of linear expansion α_a, and other steel of length l_2 having coefficient of linear expansion α_s are joined end to end. The expansion in both the rods is same on variation of temperature. Then the value of $\frac{l_1}{l_1+l_2}$ is

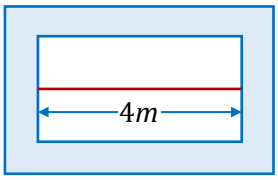
(A) $\frac{\alpha_s}{\alpha_a+\alpha_s}$ (B) $\frac{\alpha_s}{\alpha_a-\alpha_s}$ (C) $\frac{\alpha_a+\alpha_s}{\alpha_s}$ (D) None of these

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PEH102

23. A fine steel wire of length 4m is fixed rigidly in a heavy brass frame as shown in figure. It is just taut at 20°C. The tensile stress developed in steel wire if whole system is heated to 120°C is :–

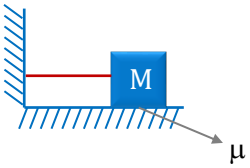
(Given $\alpha_{brass} = 1.8 \times 10^{-5}\ °C^{-1}$, $\alpha_{steel} = 1.2 \times 10^{-5}\ °C^{-1}$, $Y_{steel} = 2 \times 10^{11}\ Nm^{-2}$, $Y_{brass} = 1.7 \times 10^{7}\ Nm^{-2}$)



(A) $1.02 \times 10^4\ Nm^{-2}$ (B) $1.2 \times 10^8\ Nm^{-2}$ (C) $1.2 \times 10^6\ Nm^{-2}$ (D) $6 \times 10^8\ Nm^{-2}$

PEH105

24. A wire having cross-sectional area S is attached to wall on one side and a block of mass M on the other side which placed on a horizontal surface having coefficient of friction μ as shown. Material of wire has coefficient of thermal expansion α and Young's modulus Y. At initial temperature there is no stress in the wire. Now the wire is cooled. At what decrease in temperature the block will begin to move?



(A) $\frac{\mu Mg}{\alpha YS}$ (B) $\frac{2\mu Mg}{\alpha YS}$ (C) $\frac{\mu Mg}{2\alpha YS}$ (D) none

PEH106

For More Join: @IITJEE_Advanced

25. An aluminium sphere is heated from $0°C$ to $100°C$. The change in the volume of the sphere of radius 10 cm is ($\gamma_{Al} = 5.4 \times 10^{-5}/°C$) is

(A) $20cm^3$ (B) $12.3\ cm^3$ (C) $22.6\ cm^3$ (D) $28.9\ cm^3$

PEH107

26. The coefficient of apparent expansion of a liquid in a copper vessel is C and in a silver vessel is S. The coefficient of volume expansion of copper is γ_c. What is the coefficient of linear expansion of silver?

(A) $\frac{(C+\gamma_c+S)}{3}$ (B) $\frac{(C-\gamma_c+S)}{3}$ (C) $\frac{(C+\gamma_c-S)}{3}$ (D) $\frac{(C-\gamma_c-S)}{3}$

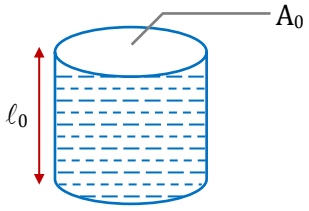
PEH110

27. Density of substance at $0°C$ is 10 g/cc and at $100°C$, its density is 9.7 g/cc. The coefficient of linear expansion of the substance will be–

(A) $10^2\ °C^{-1}$ (B) $10^{-2}\ K^{-1}$ (C) $10^{-3}\ K^{-1}$ (D) $10^{-4}\ °C^{-1}$

PEH115

28. The figure shows a glass tube (linear co-efficient of expansion is α) completely filled with a liquid of volume expansion co-efficient γ. On heating length of the liquid column does not change. Choose the correct relation between γ and α.



(A) $\gamma = \alpha$ (B) $\gamma = 2\alpha$

(C) $\gamma = 3\alpha$ (D) $\gamma = \frac{\alpha}{3}$

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PEH116

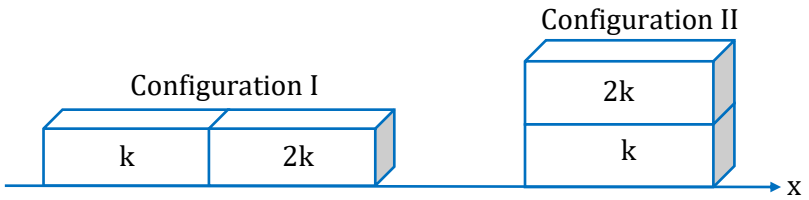
29. If two rods of length L and 2 L having coefficients of linear expansion α and 2α respectively are connected so that total length becomes 3 L, determine the average coefficient of linear expansion of the composite rod.

(A) $\frac{2\alpha}{3}$ (B) $\frac{5\alpha}{3}$ (C) $\frac{7\alpha}{3}$ (D) $\frac{8\alpha}{3}$

PEH395

Heat Transfer (Conduction)

30. Two rectangular blocks, having identical dimensions, can be arranged either in configuration I or in configuration II as shown in the figure. One of the blocks has thermal conductivity k and the other $2k$. The temperature difference between the ends along the x-axis is the same in both the configurations. It takes $9s$ to transport a certain amount of heat from the hot end to the cold end in the configuration I. The time to transport the same amount of heat in the configuration II is :-

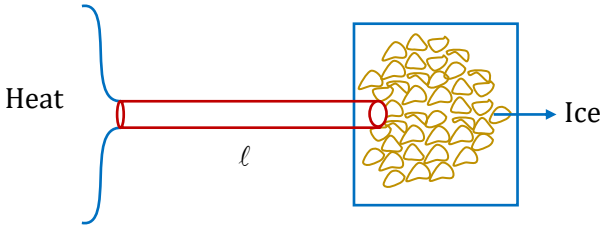


(A) 2.0 s (B) 3.0 s (C) 4.5 s (D) 6.0 s

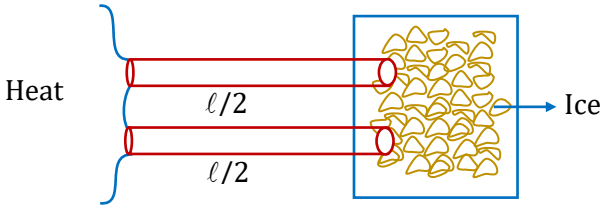
PEH117

For More Join: @IITJEE_Advanced

31. A rod of length 'λ' and cross-section 'A' is used to melt a piece of ice as shown.



Now if the rod broken into two equal parts and is arranged as shown.



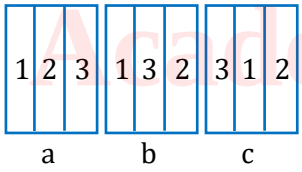
Time taken to melt ice in second case becomes.

(A) Half (B) One-forth (C) Twice (D) Four times

PEH118

32. Figure shows three different arrangements of materials 1, 2 and 3 to form a wall. Thermal conductivities are $k_1 > k_2 > k_3$. The left side of the wall is $20°C$ higher than the right side. Temperature difference ΔT across the material 1 has following relation in three cases :

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(A) $\Delta T_a > \Delta T_b > \Delta T_c$ (B) $\Delta T_a = \Delta T_b = \Delta T_c$ (C) $\Delta T_a = \Delta T_b > \Delta T_c$ (D) $\Delta T_a = \Delta T_b < \Delta T_c$

PEH120

33. One end of a $2.35m$ long and $2.0cm$ radius aluminium rod ($K = 235\ W.m^{-1}K^{-1}$) is held at $20°C$. The other end of the rod is in contact with a block of ice at its melting point. The rate in $kg.s^{-1}$ at which ice melts is [Take latent heat of fusion for ice as $\frac{10}{3} \times 10^5 J.kg^{-1}$]

(A) $48\pi \times 10^{-6}$ (B) $24\pi \times 10^{-6}$ (C) $2.4\pi \times 10^{-6}$ (D) $4.8\pi \times 10^{-6}$

PEH124

34. One end of a conducting rod is maintained at temperature $50°C$ and at the other end, ice is melting at $0°C$. The rate of melting of ice is doubled if:

(A) the temperature is made $200°C$ and the area of cross-section of the rod is doubled

(B) the temperature is made $100°C$ and length of rod is made four times

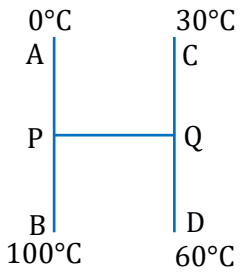
(C) area of cross-section of rod is halved and length is doubled

(D) the temperature is made $100°C$ and the area of cross-section of rod and length both are doubled.

PEH125

For More Join: @IITJEE_Advanced

35. Three identical rods *AB, CD* and *PQ* are joined as shown. *P* and *Q* are mid points of *AB* and *CD* respectively. Ends *A, B, C* and *D* are maintained at 0°*C*, 100°*C*, 30°*C* and 60°*C* respectively. The direction of heat flow in *PQ* is



(A) from *P* to *Q*

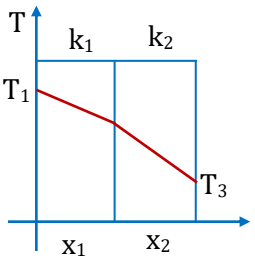
(B) from *Q* to *P*

(C) heat does not flow in *PQ*

(D) data not sufficient

PEH127

36. The temperature drop through each layer of a two layer furnace wall is shown in figure. Assume that the external temperature T_1 and T_3 are maintained constant and $T_1 > T_3$. If the thickness of the layers x_1 and x_2 are the same, which of the following statements are correct.



(A) $k_1 > k_2$

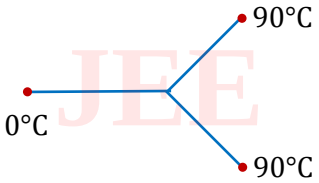
(B) $k_1 < k_2$

(C) $k_1 = k_2$ but heat flow through material (1) is larger then through (2)

(D) $k_1 = k_2$ but heat flow through material (1) is less than that through (2)

PEH128

37. Three rods made of same material and having the same cross section have been joined as shown in the figure. Each rod is of the same length. The left and right ends are kept at 0°*C* and 90°*C* respectively. The temperature of the junction of the three rods will be :



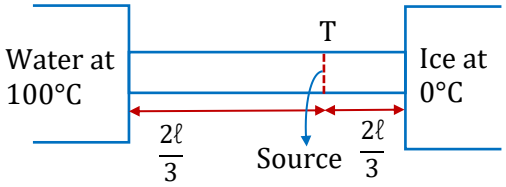
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(A) 45°*C* (B) 60°*C*

(C) 30°*C* (D) 20°*C*

PEH129

38. The rod connecting two reservoir and connected to a source is as shown in diagram. The rod is in steady state. Find the temperature of source, so that rate of melting of ice is 16 times that rate of vaporization. (Latent heat of vaporization is 540 Cal/*gm* and latent heat of fusion is 80 Cal/*gm*)



(A) 540°*C* (B) 580°*C* (C) 640°*C* (D) 740°*C*

PEH134

Heat Transfer (Radiation)

39. A body cools in 7 minutes from 60°*C* to 40°*C*. The temperature of the surrounding is 10°*C*. The temperature of the body after the next 7 minutes will be:

(A) 32°C (B) 30°C (C) 28°C (D) 34°C

PEH180

For More Join: @IITJEE_Advanced

40. A body cools from $80°C$ to $60°C$ in 5 minutes. The temperature of the surrounding is $20°C$. The time it takes to cool from $60°C$ to $40°C$ is:

(A) 500 s (2) $\frac{25}{3}$ s (3) 450 s (4) 420 s

PEH181

41. A cubical block of mass 1.0 kg and edge 5.0 cm is heated to $227°C$. It is kept in an evacuated chamber maintained at $27°C$. Assuming that the block emits radiation like a blackbody, find the rate at which the temperature of the block will decrease. Specific heat capacity of the material of the block is 400 $J/kg\text{-}K$.

(A) $0.09°C/s$ (B) $0.12°C/s$ (C) $0.15°C/s$ (D) $0.18°C/s$

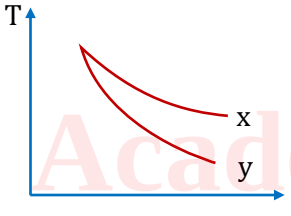
PEH163

42. A black metal foil is warmed by radiation from a small sphere at temperature 'T' and at a distance 'd'. It is found that the power received by the foil is P. If both the temperature and distance are doubled, the power received by the foil will be :

(A) 16 P (B) 4 P (C) 2 P (D) P

PEH139

43. If emissivity of bodies X and Y are e_x and e_y and absorptive power are A_x and A_y then



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(A) $e_y > e_x$; $A_y > A_x$ (B) $e_y < e_x$; $A_y < A_x$ (C) $e_y > e_x$; $A_y < A_x$ (D) $e_y = e_x$; $A_y = A_x$

PEH140

44. 'Gulab Jamuns' (assumed to be spherical) are to be heated in an oven. They are available in two sizes, one twice bigger (in radius) than the other. Pizzas (assumed to be discs) are also to be heated in oven. They are also in two sizes, one twice big (in radius) than the other. All four are put together to be heated to oven temperature. Choose the correct option from the following:

(A) Both size gulab jamuns will get heated in the same time.

(B) Smaller gulab jamuns are heated before bigger ones.

(C) Smaller pizzas are heated before bigger ones.

(D) Bigger pizzas are heated before smaller ones.

PEH141

45. Three very large plates of same area are kept parallel and close to each other. They are considered as ideal black surfaces and have very high thermal conductivity. The first and third plates are maintained at temperatures $2T$ and $3T$ respectively. The temperature of the middle (i.e. second) plate under steady state condition is :

(A) $\left(\frac{65}{2}\right)^{1/4} T$ (B) $\left(\frac{97}{4}\right)^{1/4} T$ (C) $\left(\frac{97}{2}\right)^{1/4} T$ (D) $(97)^{1/4} T$

PEH144

For More Join: @IITJEE_Advanced

46. If the temperature of the sun were to increase from T to $2T$ and its radius from R to $2R$, then the ratio of the radiant energy received on earth to what it was previously, will be-

(A) 4 (B) 16 (C) 32 (D) 64

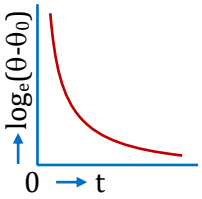
PEH146

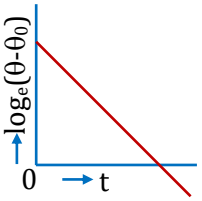
47. A system S receives heat continuously from an electrical heater of power $10W$. The temperature of S becomes constant at $50°C$ when the surrounding temperature is $20°C$. After the heater is switched off, S cools from $35.1°C$ to $34.9°C$ in 1 minute. The heat capacity of S is

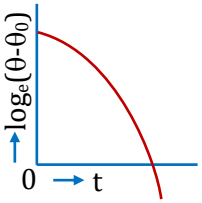
(A) $100J/°C$ (B) $300J/°C$ (C) $750J/°C$ (D) $1500J/°C$

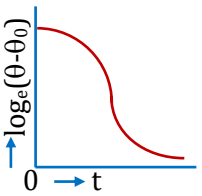
PEH150

48. A liquid in a beaker has temperature $\theta(t)$ at time t and θ_0 is temperature of surroundings, then according to Newton's law of cooling the correct graph between $\log_e(\theta - \theta_0)$ and t is :

(A) 

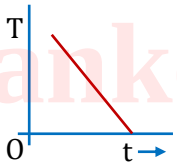
(B) 

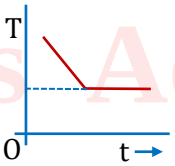
(C) 

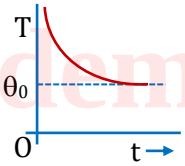
(D) 

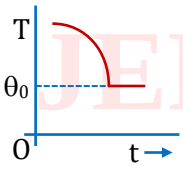
PEH155

49. If a piece of metal is heated to temperature θ and then allowed to cool in a room which is at temperature θ_0 the graph between the temperature T of the metal and time t will be closed to:

(A) 

(B) 

(C) 

(D) 

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PEH156

50. According to Wien's displacement law the wavelength of peak emission from a body is inversely proportional to the temperature of the body in kelvin. Two stars A and B with surface temperatures 3000 kelvin and 6000 kelvin respectively, are at the same distance from the Earth. They are of equal radius. The correct statement about the radiations emitted by the two stars is

(A) Both A and B will look equally bright in visible.

(B) A will look brighter than B in infrared.

(C) B will look brighter than A in infrared.

(D) A will emit more in the visible range than B in the infrared.

PEH164

KTG

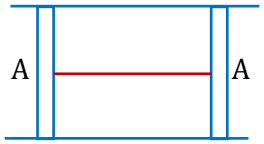
51. At a temperature T K, the pressure of $4.0g$ argon in a bulb is p. The bulb is put in a bath having temperature higher by $50K$ than the first one. $0.8g$ of argon gas had to be removed to maintained original pressure. The temperature T is equal to

(A) 510 K (B) 200 K (C) 100 K (D) 73 K

PEH222

For More Join: @IITJEE_Advanced

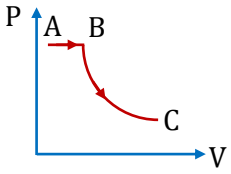
52. A cylindrical tube of cross-sectional area A has two air tight frictionless pistons at its two ends. The pistons are tied with a straight piece of metallic wire. The tube contains a gas at atmospheric pressure P_0 and temperature T_0. If temperature of the gas is doubled then the tension in the wire is

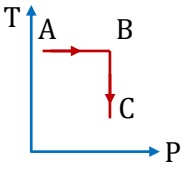


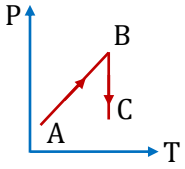
(A) $4\,P_0\,A$ (B) $P_0A/2$ (C) $P_0\,A$ (D) $2\,P_0\,A$

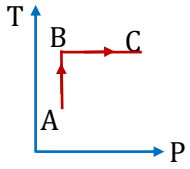
PEH223

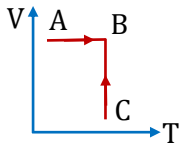
53. A process is shown in the diagram. Which of the following curves may represent the same process?



(A) 

(B) 

(C) 

(D) 

PEH224

54. An open and wide glass tube is immersed vertically in mercury in such a way that length 0.05 m extends above mercury level. The open end of the tube is closed and the tube is raised further by 0.43m. The length of air column above mercury level in the tube will be

(Take P_{atm} = 76 cm of mercury)

(A) 0.215 cm (B) 0.2 cm (C) 0.135 cm (D) 0.4 cm

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PEH225

55. The value of the ratio C_p/C_v for hydrogen is 1.67 at 30 K but decreases to 1.4 at 300 K as more degrees of freedom become active. During this rise in temperature (assume H_2 as ideal gas),

(A) C_p remains constant but C_v increases

(B) C_p decreases but C_v increases

(C) Both C_p and C_v decrease by the same amount

(D) Both C_p and C_v increase by the same amount

PEH226

56. In a constant volume gas thermometer, the pressure of the working gas is measured by the difference in the levels of mercury in the two arms of a U-tube connected the gas at one end. When the bulb is placed at the room temperature 27.0°C, the mercury column in the arm open to atmosphere stands 5.00 cm above the level of mercury in the other arm. When the bulb is placed in a hot liquid, the difference of mercury levels becomes 45.0 cm. Calculate the temperature of the liquid. (atmospheric pressure = 75.0 cm of mercury.)

(A) 450°C (B) 177°C (C) 40.5°C (D) 243 °C

PEH227

57. The ratio of speed of sound in hydrogen gas to the speed of sound in oxygen gas at the same temperature is:

(A) 4 : 1 (B) 1 : 2 (C) 1 : 4 (D) 1 : 1

PEH229

For More Join: @IITJEE_Advanced

58. One mole of an ideal monoatomic gas $\left(\gamma=\frac{5}{3}\right)$ is mixed with one mole of a diatomic gas $\left(\gamma=\frac{7}{5}\right)$. ($\gamma$ denotes the ratio of specific heat at constant pressure, to that at constant volume) find γ for the mixture ?

(A) $\frac{3}{2}$ (B) $\frac{5}{2}$ (C) $\frac{7}{2}$ (D) 3

PEH396

Word done by Gas and First Law of Thermodynamics

59. Two monoatomic ideal gas at temperature T_1 and T_2 are mixed. There is no loss of energy. If the masses of molecules of the two gases are m_1 and m_2 and number of their molecules are n_1 and n_2 respectively. The temperature of the mixture will be:

(A) $\frac{T_1+T_2}{n_1+n_2}$ (B) $\frac{T_1}{n_1}+\frac{T_2}{n_2}$ (C) $\frac{n_2T_1+n_1T_2}{n_1+n_2}$ (D) $\frac{n_1T_1+n_2T_2}{n_1+n_2}$

PEH230

60. The value of $C_P - C_v$ is 1.09 R for a gas sample in state A and is 1.00 R in state B. Let T_A, T_B denote the temperature and P_A and P_B denote the pressure of the states A and B respectively. Then

(A) $P_A < P_B$ and $T_A > T_B$ (B) $P_A > P_B$ and $T_A > T_B$

(C) $P_A = P_B$ and $T_A < T_B$ (D) $P_A > P_B$ and $T_A < T_B$

PEH231

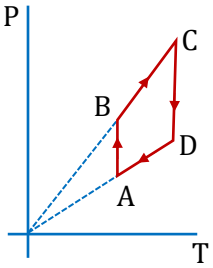
61. An ideal gas expands isothermally from a volume V_1 to V_2 and then compressed to original volume V_1 adiabatically. Initial pressure is P_1 and final pressure is P_3. The total work done is W. Then

(A) $P_3 > P_1$, $W > 0$ (B) $P_3 < P_1$, $W < 0$ (C) $P_3 > P_1$, $W < 0$ (D) $P_3 = P_1$, $W = 0$

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PEH232

62. Pressure versus temperature graph of an ideal gas is shown in figure



(A) During the process AB work done by the gas is positive

(B) During the process CD work done by the gas is negative

(C) During the process BC internal energy of the gas is increasing

(D) None

PEH233

63. Water is heated in an open pan where the air pressure is 10^{+5} Pa. The water remains a liquid, which expands by a small amount as it is heated. Determine the ratio of the heat absorbed by the water to the work done by water. (γ for water = $10^{-3}/C$, S = 1 cal/$gm°C$)

(A) 4.2×10^3 (B) 4.2×10^5 (C) 4.2×10^2 (D) 4.2×10^4

PEH234

For More Join: @IITJEE_Advanced

64. A vertical cylinder with heat-conducting walls is closed at the bottom and is fitted with a smooth light piston. It contains one mole of an ideal gas. The temperature of the gas is always equal to the surrounding's temperature, T_0. The piston is moved up slowly to increase the volume of the gas to η times. Which of the following is incorrect?

(A) Work done by the gas is $RT_0 \ln \eta$.

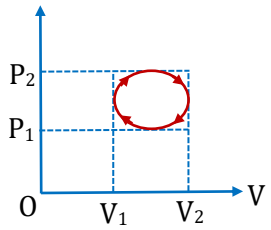
(B) Work done against the atmosphere is $RT_0(\eta - 1)$.

(C) There is no change in the internal energy of the gas.

(D) The final pressure of the gas is $\frac{1}{(\eta-1)}$ times its initial pressure.

PEH235

65. In a cyclic process shown on the $P - V$ diagram the magnitude of the work done is :



(A) $\pi\left(\frac{P_2 - P_1}{2}\right)^2$ (B) $\pi\left(\frac{V_2 - V_1}{2}\right)^2$

(C) $\frac{\pi}{4}(P_2 - P_1)(V_2 - V_1)$ (D) $\pi(P_2V_2 - P_1V_1)$

PEH236

66. A polyatomic gas with six degrees of freedom does $25J$ of work when it is expanded at constant pressure. The heat given to the gas is :-

(A) $100J$ (B) $150J$ (C) $200J$ (D) $250J$

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PEH237

67. 2 moles of a monoatomic gas are expanded to double its initial volume, through a process P/V = constant. If its initial temperature is 300 K, then which of the following is not true.

(A) $\Delta T = 900\ K$ (B) $\Delta Q = 3200\ R$ (C) $\Delta Q = 3600\ R$ (D) $W = 900\ R$

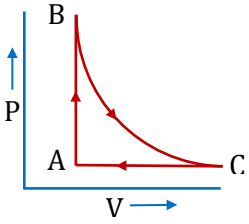
PEH238

68. One mole of an ideal gas at temperature T_1 expends according to the law $\frac{P}{V^2} = a$ (constant). The work done by the gas till temperature of gas becomes T_2 is :

(A) $\frac{1}{2}R(T_2 - T_1)$ (B) $\frac{1}{3}R(T_2 - T_1)$ (C) $\frac{1}{4}R(T_2 - T_1)$ (D) $\frac{1}{5}R(T_2 - T_1)$

PEH239

69. One mole of monoatomic ideal gas undergoes a cyclic process $ABCA$ as shown in figure. Process BC is adiabatic. The temperatures at A, B and C are 300, 600 and 450K respectively. Choose the correct statement(s).



(A) In process CA change in internal energy is $225R$.

(B) In process AB change in internal energy is $-150R$.

(C) In process BC change in internal energy is $-225R$.

(D) Change in internal energy during the whole cyclic process is $+150R$.

PEH240

For More Join: @IITJEE_Advanced

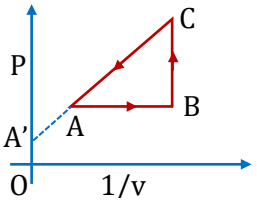
70. An enclosed ideal gas is taken through a cycle as shown in the figure. Then

(A) Along *AB*, temperature decreases while along *BC* temperature increases

(B) Along *AB*, temperature increases while along *BC* the temperature decreases.

(C) Along *CA* work is done by the gas and the internal energy remains constant.

(D) Along *CA* work is done on the gas and internal energy of the gas increases.

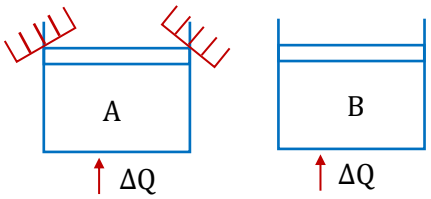


PEH241

71. Two identical vessels *A* and *B* contain equal amount of ideal monoatomic gas. The piston of *A* is fixed but that of *B* is free. Same amount of heat is absorbed by *A* and *B*. If *B's* internal energy increases by 100 *J* the change in internal energy of *A* is :-

(A) 100 *J* (B) $\frac{500}{3}J$

(C) 250 *J* (D) none of these



PEH242

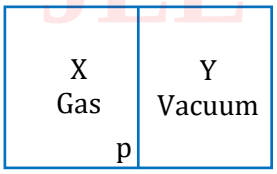
72. A closed container is fully insulated from outside. One half of it is filled with an ideal gas *X* separated by a plate *P* from the other half *Y* which contains a vacuum as shown in figure. When *P* is removed, *X* moves into *Y*. Which of the following statements is correct?

(A) No work is done by *X*

(B) *X* decreases in temperature

(C) *X* increases in internal energy

(D) *X* doubles in pressure

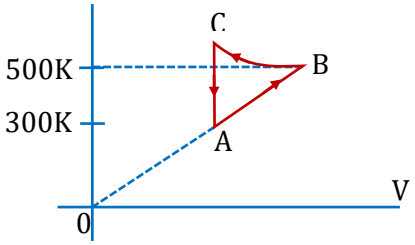


PEH243

73. Consider the cyclic process *ABCA*, shown in figure, performed on a sample of 2.0 mole of an ideal gas. A total of 1200 *J* of heat is withdrawn from the sample in the process. Find the work done by the gas during the part *BC*.

(A) –4480 *J* (B) –4520 *J*

(C) –4540 *J* (D) –4460 *J*



PEH244

Second Law of Thermodynamics

74. A Carnot engine whose source temperature is at 400 *K* takes 100 *kcal* of heat at this temperature in each cycle and gives 70 *kcal* to the sink. Calculate,

(i) the temperature of the sink and (ii) the efficiency of the engine.

(A) 280 *K*, 30 % (B) 270 *K*, 30% (C) 280 *K*, 10 % (D) None of these

PEH245

For More Join: @IITJEE_Advanced

75. A Carnot engine takes 3×10^6 *cal* of heat from reservoir at 627° and gives it to a sink at 27°*C*. Then work done by the engine is :-
(A) 4.2×10^6 *J* (B) 8.4×10^6 *J* (C) 16.8×10^6 *J* (D) zero

PEH246

76. A refrigerator freezes 5 *kg* of water at 0°*C* into ice at 0°*C* in a time interval of 20 minutes. Assume that room temperature is 20°*C*. Calculate the minimum power needed to accomplish it ?
(A) 24.4 Watt (B) 0.1025 Watt (C) 0.0244 Watt (D) 102.5 Watt

PEH247

77. An ideal refrigerator operates according to the reverse Carnot cycle and transmits heat from a cold source with water at a temperature of 0°*C* to a boiler with water at a temperature of 100°*C*. What amount of water must be frozen in the cooler to convert 1 *kg* of water into vapour in the boiler?
(A) 4.94 *kg* (B) 3.24 *kg* (C) 5.63 *kg* (D) 2.12 *kg*

PEH248

78. According to the second law of thermodynamics:
(A) Heat energy cannot be completely converted to work
(B) Work cannot be completely converted to heat energy
(C) For all cyclic processes we have $dQ/T < 0$
(D) The reason all heat engine efficiencies are less than 100% is friction, which is unavoidable

PEH249

79. The change in entropy between the two adiabatic lines of a Carnot cycle is 1 *kcal*/*deg*. The temperature difference between the two isothermal lines is equal to 100°*C*. What amount of heat will be converted into work during this cycle?
(A) 420 *kJ* (B) 840 *kJ* (C) 210 *kJ* (D) 105 *kJ*

PEH250

Entropy

Rankers Academy JEE

80. Which one of the following pairs of equations describes an irreversible heat engine?
(A) $\oint \delta Q > 0$ and $\oint \frac{\delta Q}{T} < 0$ (B) $\oint \delta Q < 0$ and $\oint \frac{\delta Q}{T} < 0$
(C) $\oint \delta Q > 0$ and $\oint \frac{\delta Q}{T} > 0$ (D) $\oint \delta Q < 0$ and $\oint \frac{\delta Q}{T} > 0$

PEH251

81. Oxygen (10 grammes) is heated from t_1 = 50°*C* to t_2 = 150°*C*. Find the change in entropy if the oxygen is heated: (1) isochorically, (2) isobarically.
(A) $\Delta S_V = 1.76\ J/K$; $\Delta S_P = 2.46\ J/K$ (B) $\Delta S_V = \Delta S_P = 1.76\ J/K$
(C) $\Delta S_V = \Delta S_P = 2.46\ J/K$ (D) $\Delta S_V = 2.46\ J/K$; $\Delta S_P = 1.76\ J/K$

PEH252

82. An ideal gas expands into a vacuum in a rigid vessel. As a result there is :
(A) a change in entropy (B) an increase of pressure
(C) a change in temperature (D) a decrease of internal energy

PEH253

83. A metallic object cools from temperature T_1 to temperature T_2. The specific heat of object varies as $C = a + bT^2$ (where T is in kelvin). The change in specific entropy of the object during the cooling process is :
(A) $a\ln\left(\frac{T_2}{T_1}\right)+0.5b\left(T_2^2-T_1^2\right)$ (B) $a\ln\left(\frac{T_2}{T_1}\right)$
(C) $a\ln\left(T_1+T_2\right)+0.5b\left(T_2^2-T_1^2\right)$ (D) $a\ln\left[a+b\left(\frac{T_1+T_2}{2}\right)\right]$

PEH254

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. 50 *gm* ice at – 10°*C* is mixed with 20 *gm* steam at 100°*C*. When the mixture finally reaches its steady state inside a calorimeter of water equivalent 1.5 *gm* then :

[Assume calorimeter was initially at 0°*C*, Take latent heat of vaporization of water = 540 *cal/gm*, Latent heat of fusion of water = 80 *cal/gm*, specific heat capacity of water = 1 *cal/gm°C*, specific heat capacity of ice = 0.5 *cal/gm°C*]

(A) Mass of water remaining is : 67.4 *gm*

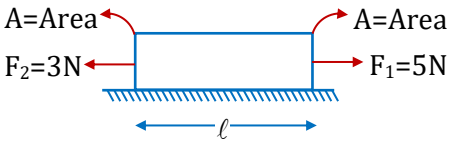
(B) Mass of steam remaining is : 2.6 *gm*

(C) Mass of water remaining is : 67.87 *gm*

(D) Mass of steam remaining is : 2.13 *gm*

PEH073

2. A block of mass *M*, area of cross-section *A* and length ℓ is placed on smooth horizontal floor. Two forces $F_1 = 5N$ and $F_2 = 3\ N$ are applied on the block as shown in the figure. If *Y* is Young's modulus of elasticity of material, then total extension in the length of the block will be mark **INCORRECT**



(A) $\frac{8\ell}{AY}$ (B) $\frac{2\ell}{AY}$ (C) $\frac{4\ell}{AY}$ (D) $\frac{5\ell}{2AY}$

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PEH397

3. A bimetallic strip is formed out of two identical strips one of copper and the other of brass. The coefficient of linear expansion of the two metals are α_C and α_B. On heating, the temperature of the strip goes up by ΔT and the strip bends to form an arc of radius of curvature *R*. Then *R* is

(A) proportional at ΔT

(B) inversely proportional to ΔT

(C) proportional to $|\alpha_B - \alpha_C|$

(D) inversely proportional to $|\alpha_B - \alpha_C|$

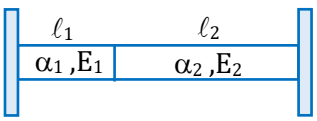
PEH113

4. Two rods of different materials are placed between massive walls as shown in figure. The cross section of the rods is *A* their moduli of elastically are E_1 and E_2 respectively. If rods are heated by *T* degrees, then (coefficients of linear expansion of material of rods are α_1 and α_2 respectively)

(A) the strain produced in both rods is equal.

(B) the stress produced in both the rods is equal.

(C) the force exerted by one rod on the other is $\frac{\alpha_1\ell_1 + \alpha_2\ell_2}{\frac{\ell_1}{E_1} + \frac{\ell_2}{E_2}} AT$.

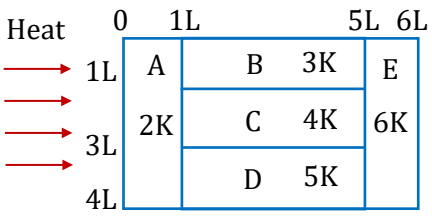


(D) the force exerted by one rod on the other is $\frac{(E_1 - E_2)^2 A}{E_1 + E_2}$.

PEH114

For More Join: @IITJEE_Advanced

5. A composite block is made of slabs A, B, C, D and E of different thermal conductivities (given in terms of a constant K) and sizes (given in terms of length, L) as shown in the figure. All slabs are of same width. Heat 'Q' flows only from left to right through the blocks. Then in steady state



(A) heat flow through A and E slabs are same

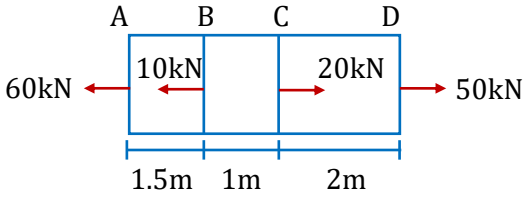
(B) heat flow through slab E is maximum

(C) temperature difference across slab E is smallest

(D) heat flow through C = heat flow through B + heat flow through D

PEH131

6. A Steel rod of cross-sectional area $1m^2$ is acted upon by forces as shown in the Fig. Determine the total elongation of the bar. Take $Y = 2.0 \times 10^{11} N/m^2$. Choose incorrect options

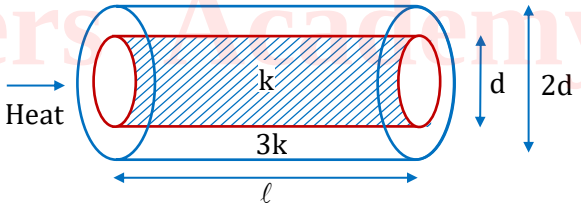


(A) 1.3 μm (B) 1.7 μm (C) 0.9 μm (D) 1.2 μm

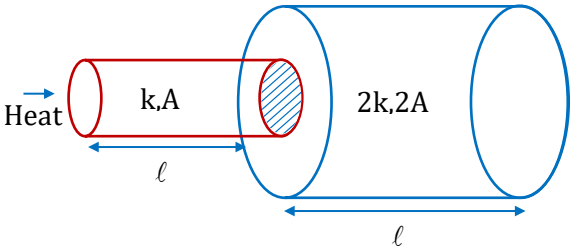
PEH398

7. Different combination of cylinder are shown in the figure. Select the correct alternatives for the thermal resistance of the following combinations :

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A thin cylinder of diameter 'd' is surrounded by another cylinder.



A : Area of cross-section
k : Thermal conductivity
ℓ : Length of cylinder
d : diameter of cylinder

(A) Thermal resistance of '1' combination is $\frac{2\ell}{5k\pi d^2}$

(B) Thermal resistance of '2' combination is $\frac{4\ell}{5kA}$

(C) Thermal resistance of '1' combination is $\frac{5\ell}{2k\pi d^2}$

(D) Thermal resistance of '2' combination is $\frac{5\ell}{4kA}$

PEH132

For More Join: @IITJEE_Advanced

8. A metal cylinder of mass 0.5 kg is heated electrically by a 12 W heater in a room at 15°C. The cylinder temperature rises to 25°C in 5 min and finally becomes constant at 45°C. Assuming that the rate of heat loss is proportional to the excess temperature over the surroundings

(A) The rate of loss of heat of the cylinder to surrounding at 20°C is 2 W

(B) The rate of loss of heat of the cylinder to surrounding at 45°C is 2 W

(C) Specific heat capacity of metal is $\frac{240}{\ln(3/2)} J/kg°C$

(D) None of these

PEH165

9. Let v_{av}, v_{rms} and v_p respectively denote mean speed, root mean square speed and the most probable speed of the molecule in an ideal monoatomic gas at absolute temperature T. The mass of a molecule is m then :

(A) no molecule can have speed greater than v_{rms}

(B) no molecule can have speed less than $v_p/\sqrt{2}$

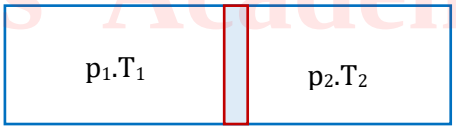
(C) $v_p < v_{av} < v_{rms}$

(D) the average kinetic energy of a molecule is $3/4\ mv_p^2$

PEH255

10. Figure shows an adiabatic cylindrical tube of volume V_0 divided in two parts by a frictionless adiabatic separator. Initially, the separator is kept in the middle, an ideal gas at pressure p_1 and temperature T_1 is injected into the left part and another ideal gas at pressure p_2 and temperature T_2 is injected into the right part. $\frac{C_p}{C_v} = \gamma$ is the same for both the gases. The separator is slid slowly and is released at a position where it can stay in equilibrium. Find

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(A) The volumes of the two parts are $\frac{p_1^{1/\gamma}V_0}{p_1^{1/\gamma}+p_2^{1/\gamma}}$ and $\frac{p_2^{1/\gamma}V_0}{p_1^{1/\gamma}+p_2^{1/\gamma}}$ respectively.

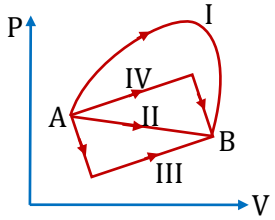
(B) The volumes of the two parts are $\frac{p_2^{1/\gamma}V_0}{p_1^{1/\gamma}+p_2^{1/\gamma}}$ and $\frac{p_1^{1/\gamma}V_0}{p_1^{1/\gamma}+p_2^{1/\gamma}}$ respectively.

(C) The heat given to the gas in the left part is zero.

(D) The final common pressure of the gases is $\left(\frac{p_1^{1/\gamma}+p_2^{1/\gamma}}{2}\right)^{\gamma}$

PEH256

11. Figure shows the P-V diagram of an ideal gas undergoing a change of state from A to B. Four different parts I, II, III and IV as shown in the figure may lead to the same change of state.



(A) Change in internal energy is same in IV and III cases, but not in I and II.

(B) Change in internal energy is same in all the four cases.

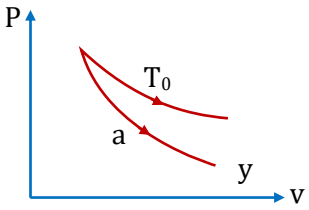
(C) Work done is maximum in case I

(D) Work done is minimum in case II.

PEH257

For More Join: @IITJEE_Advanced

12. Two gases have the same initial pressure, volume and temperature. They expand to the same final volume, one adiabatically and the other isothermally



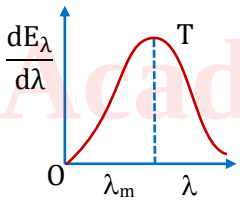
(A) The final temperature is greater for the isothermal process
(B) The final pressure is greater for the isothermal process
(C) The work done by the gas is greater for the isothermal process
(D) All the above options are incorrect

PEH258

COMPREHENSION TYPE QUESTIONS

PARAGRAPH FOR QUESTION NO. 13 AND 14

The figure shows a radiant energy spectrum graph for a black body at a temperature *T*.



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13. Choose the **CORRECT** statement(s)

(A) The radiant energy is not equally distributed among all the possible wavelengths
(B) For a particular wavelength the spectral intensity is maximum
(C) The area under the curve is equal to the total rate at which heat is radiated by the body at that temperature
(D) None of these

PEH151

14. If the temperature of the body is raised to a higher temperature *T'*, then choose the **CORRECT** statement(s)

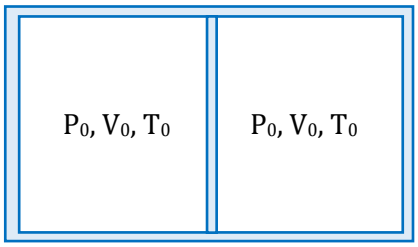
(A) The intensity of radiation for every wavelength increases
(B) The maximum intensity occurs at a shorter wavelength
(C) The area under the graph increases
(D) The area under the graph is proportional to the fourth power of temperature

PEH152

For More Join: @IITJEE_Advanced

PARAGRAPH FOR QUESTION NO. 15 TO 17

One mole of a monoatomic ideal gas occupies two chambers of a cylinder partitioned by means of a movable piston. The walls of the cylinder as well as the piston are thermal insulators. Initially equal amounts of gas fill both the chambers at (P_0, V_0, T_0). A coil is burnt in the left chamber which absorbs heat and expands, pushing the piston to the right. The gas on the right chamber is compressed until the pressure becomes 32 P_0.



15. The final volume of left chamber is :

(A) $\frac{V_0}{8}$ (B) $\frac{15}{8}V_0$ (C) $\frac{7}{8}V_0$ (D) $\frac{9}{8}V_0$

PEH259

16. The work done on the gas in the right chamber is :

(A) $\frac{9}{2}P_0V_0$ (B) $-\frac{9}{2}P_0V_0$ (C) $\frac{13}{2}P_0V_0$ (D) $\frac{17}{2}P_0V_0$

PEH260

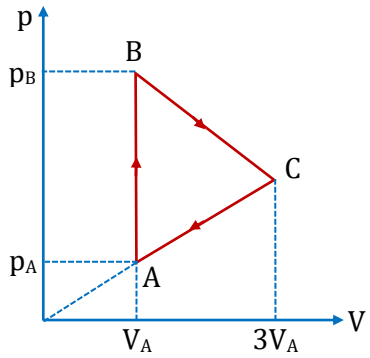
17. The change in internal energy of the gas in the left chamber is :

(A) $\frac{186}{4}RT_0$ (B) $\frac{177}{4}RT_0$ (C) $\frac{59}{2}RT_0$ (D) $\frac{131}{4}RT_0$

PEH261

PARAGRAPH FOR QUESTION NO. 18 AND 19

A sample of ideal gas is taken through the cyclic process shown in the figure. The temperature of the gas at state A is T_A = 200 K. At states B and C the temperature of the gas is the same.

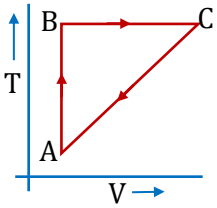


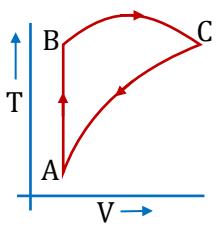
18. Net work done by the gas in the process is

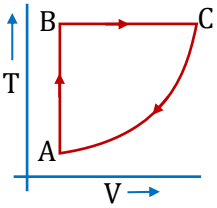
(A) 2 P_AV_A (B) 4 P_AV_A (C) 6 P_AV_A (D) 8 P_AV_A

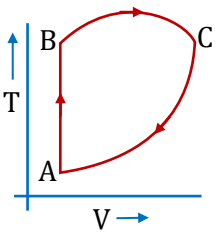
PEH262

19. Which of the following graphs best represent the cyclic process in T-V diagram.

(A) 

(B) 

(C) 

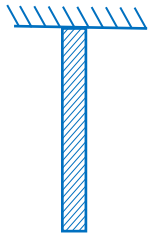
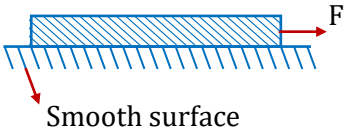
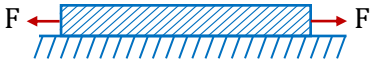
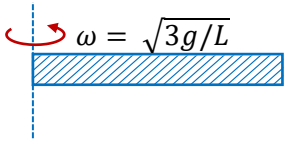
(D) 

PEH263

For More Join: @IITJEE_Advanced

MATRIX MATCH TYPE QUESTION

20. In Column-I different configuration of uniform rod of mass *m*, length '*L*', cross-sectional area '*A*' and Young's modulus of elasticity '*Y*' are as shown. Match the column-I with column–II.

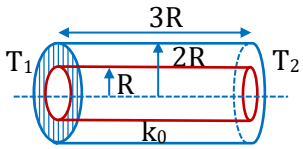
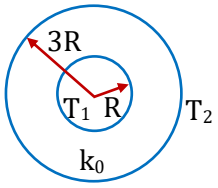
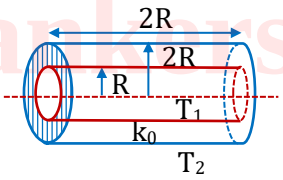
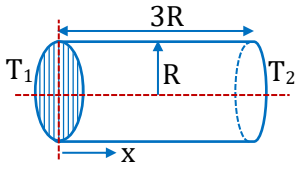
	Column–I		**Column–II**
(A)	 A rod is hanging vertically with fixed support	(P)	Total elongation in the rod is $\frac{mgL}{2AY}$
(B)	 A constant force $F = mg$ is applied at one end of the rod.	(Q)	Total elongation in the rod is $\frac{mgL}{AY}$
(C)	 Equal and opposite forces ($F = mg$) are applied at both end of the rod	(R)	Uniform stresses developed in the rod.
(D)	 A rod is rotating with constant angular velocity about end point in gravity free space.	(S)	Non-uniform stress developed in the rod.
		(T)	Elastic potential energy stored in the rod is $\frac{m^2g^2L}{6AY}$

PEH096

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For More Join: @IITJEE_Advanced

21. Entries in column-I consists of diagrams of thermal conductors. The type of conductor and direction of heat flows are listed below. Entries in column-II consists of the magnitude of rate of heat flow belonging to any of the entries in column-I. If temperature difference in all the cases is $(T_1 - T_2)$, then match column.

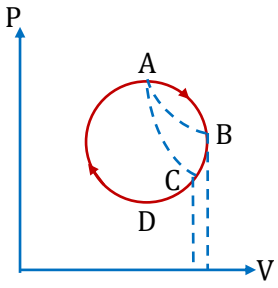
Column-I		**Column-II**	
(A)		(P)	$6\pi k_0 R(T_1 - T_2)$
(B)	 Thick spherical shell, radial flow	(Q)	$\frac{\pi k_0 R}{3\ln 2}(T_1 - T_2)$
(C)	 Thick cylindrical shell, radial flow	(R)	$\pi k_0 R(T_1 - T_2)$
(D)	 Thick cylindrical shell, radial flow	(S)	$\frac{4\pi k_0 R}{\ln 2}(T_1 - T_2)$

PEH135

Rankers Academy JEE

For More Join: @IITJEE_Advanced

22. For an ideal gas a process *PV* diagram is a circle. An adiabatic from *A* passes through *C*. An isotherm from *A* passes through *B*. We take a part of the circular cyclic process. Comment on the sign of the quantity of column-I.



	Column-I		**Column-II**
(A)	Heat given to the gas in going from *A* to *C* along circle	(P)	Positive
(B)	Heat given to the gas in going from *B* to *C* along circle	(Q)	Negative
(C)	Heat given to the gas in going from *C* to *D* along circle	(R)	Zero
		(S)	Can't be said

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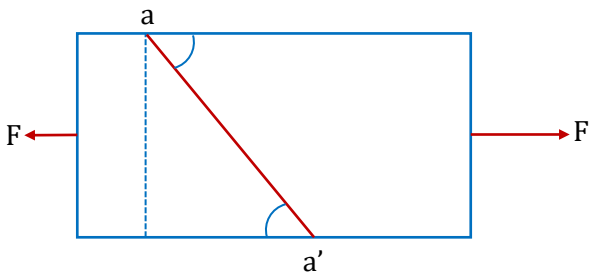
PEH264

For More Join: @IITJEE_Advanced

EXERCISE (S)

Elasticity

1. Consider a long steel bar under a tensile stress due to forces $\vec{F}$ acting at the edges along the length of the bar (Fig.). Consider a plane making an angle θ with the length. What are the tensile and shearing stresses on this plane?



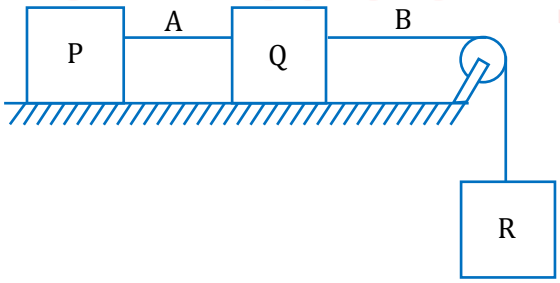
(a) For what angle is the tensile stress a maximum?

(b) For what angle is the shearing stress a maximum?

PEH079

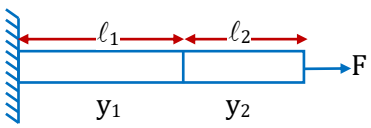
2. Each of the three blocks P, Q and R shown in figure has a mass of 3 kg. Each of the wires A and B has cross-sectional area 0.005 cm^2 and Young's modulus 2×10^{11} N/m^2. Neglect friction. Find the longitudinal strain developed in each of the wires. Take $g = 10$ m/s^2.

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PEH166

3. Find the



(i) Net elongation of composite rod approximately. ($x \times 10^{-11} m$) then x =

(ii) Y_{eq} of the composite rod ($x \times 10^{11} N/m^2$)

(Assume A = Area of cross section of each rod), then x =

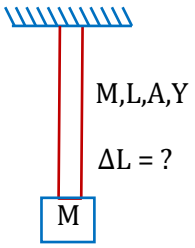
$\ell_1 = \ell_2 = 1m$, $F = 2N/m^2$, $A = 1sq.m^2$

$y_1 = 2 \times 10^{11}$ N/m^2, $y_2 = 3 \times 10^{11}$ N/m^2

PEH167

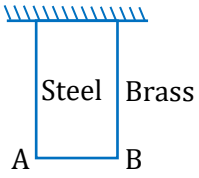
For More Join: @IITJEE_Advanced

4. Let a rope of mass M and length L and young's modulus of elasticity Y and cross-sectional area A is hung vertically. A mass M is attached to its lower end. Calculate the increase in length.



PEH168

5. A light rigid bar AB is suspended horizontally from two vertical wires, one of steel and one of brass, as shown in figure. Each wire is 2.00 m long. The diameter of the steel wire is 0.60 mm and the length of the bar AB is 0.20 m. When a mass of 10 kg is suspended from the centre of AB bar remains horizontal.



(i) What is the tension in each wire?

(ii) Calculate the extension of the steel wire and the energy stored in it.

(iii) Calculate the diameter of the brass wire.

(iv) If the brass wire were replaced by another brass wire of diameter 1 mm, where should the mass be suspended so that AB would remain horizontal? The Young modulus for steel = 2.0×10^{11} Pa, the Young modulus for brass = 1.0×10^{11} Pa.

PEH086

Rankers Academy JEE

Calorimetry

6. A copper calorimeter of mass 100 gm contains 200 gm of a mixture of ice and water. Steam at 100°C under normal pressure is passed into the calorimeter and the temperature of the mixture is allowed to rise to 50°C. If the mass of the calorimeter and its contents is now 330 gm, what was the ratio of ice and water in the beginning? Neglect heat losses.

Given : Specific heat capacity of copper = 0.42×10^3 $J\,kg^{-1}K^{-1}$,

Specific heat capacity of water = 4.2×10^3 $J\,kg^{-1}K^{-1}$,

Specific heat of fusion of ice = 3.36×10^5 $J\,kg^{-1}$

Latent heat of condensation of steam = 22.5×10^5 Jkg^{-1}

PEH169

Thermal Expansion

7. A clock pendulum made of invar has a period of 0.5 sec at 20°C. If the clock is used in a climate where average temperature is 30°C, approximately. How much fast or slow will the clock run in 10^6 sec. ($\alpha_{invar} = 1 \times 10^{-6}/°C$)

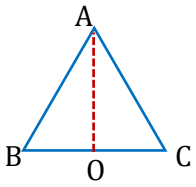
PEH103

8. A steel ball initially at a pressure of 1.0×10^5 Pa is heated from 20°C to 120°C keeping its volume constant. Find the pressure inside the ball. Coefficient of linear expansion of steel = $12 \times 10^{-6}/°C$ and bulk modulus of steel = 1.6×10^{11} N/m^2.

PEH170

For More Join: @IITJEE_Advanced

9. Three aluminium rods of equal length form an equilateral triangle ABC. Taking O (mid point of rod BC) as the origin. Find the increase in Y-coordinate of center of mass per unit change in temperature of the system. Assume the length of the each rod is $2m$, and $\alpha_{al} = 4\sqrt{3}\times10^{-6}/°C$.



PEH171

10. Two rods made of different materials are placed between massive walls. The cross section of the rods are A_1 and A_2, their lengths L_1 and L_2 coefficients of linear expansion α_1 and α_2 and the moduli of elasticity of their materials Y_1 and Y_2 respectively. If the rods are heated by $t°C$, find the force F with which the rods act on each other.

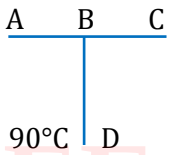
PEH211

11. A rod of length $2m$ at $0°C$ and having expansion coefficient $\alpha = (3x + 2) \times 10^{-6}\,°C^{-1}$ where x is the distance (in cm) from one end of rod. The length of rod at $20°C$ is :

PEH399

Heat Transfer (Conduction)

12. Three conducting rods of same material and cross-section are shown in figure. Temperature of A, D and C are maintained at $20°C$, $90°C$ and $0°C$. Find the ratio of length BD and BC if there is no heat flow in AB.



PEH122

13. A lagged stick of cross section area 1 cm^2 and length 1 m is initially at a temperature of $0°C$. It is then kept between 2 reservoirs of temperature $100°C$ and $0°C$. Specific heat capacity is 10 $J/kg°C$ and linear mass density is 2 kg/m. Find:

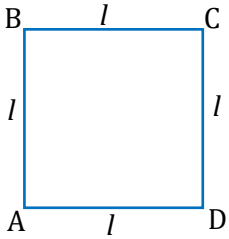


(a) temperature gradient along the rod in steady state.

(b) total heat absorbed by the rod to reach steady state.

PEH123

14. In the square frame of side l of metallic rods, the corners A and C are maintained at T_1 and T_2 respectively. The rate of heat flow from A to C is ω. If A and D are instead maintained T_1 and T_2 respectively. Find the total rate of heat flow.



PEH126

For More Join: @IITJEE_Advanced

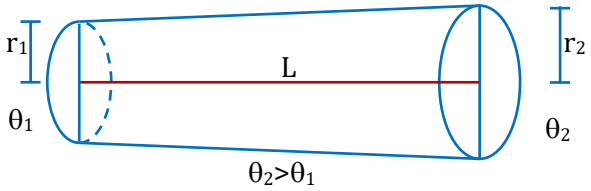
15. A metal rod AB of length $10x$ has its one end A in ice at $0°C$ and the other end B in water at $100°C$. If a point P on the rod is maintained at $400°C$, then it is found that equal amounts of water and ice evaporate and melt per unit time. The latent heat of evaporation of water is 540 cal/g and latent heat of melting of ice is 80 cal/g. If the point P is at a distance of λx from the ice end A, find the value of λ. [Neglect any heat loss to the surrounding]

PEH130

16. One end of copper rod of uniform cross-section and of length 1.5 meters is in contact with melting ice and the other end with boiling water. At what point along its length should a temperature of $200°C$ be maintained, so that in steady state, the mass of ice melting is equal to that of steam produced in the same interval of time? Assume that the whole system is insulated from the surroundings.

PEH133

17. Find the rate of heat flow through a cross section of the rod shown in figure($\theta_2 > \theta_1$). Thermal conductivity of the material of the rod is K



PEH212

18. Two bodies of masses m_1 and m_2 and specific heat capacities s_1 and s_2 are connected by a rod of length l, cross-sectional area A, thermal conductivity K and negligible heat capacity. The whole system is thermally insulated. At time $t = 0$, the temperature of the first body is T_1 and the temperature of the second body is T_2 ($T_2 > T_1$). Find the temperature difference between the two bodies at time t.

Pankaj Academy JEE

PEH213

19. Figure shows two adiabatic vessels, each containing a mass m of water at different temperatures. The ends of a metal rod of length L, area of cross section A and thermal conductivity K, are inserted in the water as shown in the figure. Find the time taken for the difference between the temperatures in the vessels to become half of the original value. The specific heat capacity of water is s. Neglect the heat capacity of the rod and the container and any loss of heat to the atmosphere.



PEH183

Heat Transfer (Radiation)

20. A hollow metallic sphere of radius 20 cm surrounds a concentric metallic sphere of radius 5 cm. The space between the two spheres is filled with a non-metallic material. The inner and outer spheres are maintained at $50°C$ and $10°C$ respectively and it is found that 100 J of heat passes from the inner sphere to the outer sphere per second. Find the thermal conductivity of the material between the spheres.

PEH210

For More Join: @IITJEE_Advanced

21. A cylindrical block of length 0.4 m an area of cross-section $0.04m^2$ is placed coaxially on a thin metal disc of mass 0.4 kg and of the same cross-section. The upper face of the cylinder is maintained at a constant temperature of $400K$ and the initial temperature of the disc is $300K$. If the thermal conductivity of the material of the cylinder is 10 watt/m-K and the specific heat of the material of the disc is 600 J/kg-K, how long will it take for the temperature of the disc to increase to $350K$? Assume, for purposes of calculation, the thermal conductivity of the disc to be very high and the system to be thermally insulated except for the upper face of the cylinder.

PEH142

22. Two spheres of same radius R have their densities in the ratio 8 : 1 and the ratio of their specific heats are 1 : 4. If by radiation their rates of fall of temperature are same, then find the ratio of their rates of losing heat.

PEH143

23. Two spherical bodies A (radius 6 cm) and B (radius 18 cm) are at temperatures T_1 and T_2, respectively. The maximum intensity in the emission spectrum of A is at 500 nm and in that of B is at 1500 nm. Considering then to be black bodies, what will be the ratio of the rate of total energy radiated by A to that of B?

PEH147

24. A solid receives heat by radiation over its surface at the rate of 4 kW. The heat convection rate from the surface of solid to the surrounding is 5.2 kW, and heat is generated at a rate of 1.7 kW over the volume of the solid. The rate of change of the average temperature of the solid is 0.5 $°Cs^{-1}$. Find the heat capacity of the solid.

PEH153

Rankers Academy JEE

25. Assume that the total surface area of a human body is 1.6 m^2 and that it radiates like an ideal radiator. Calculate the amount of energy radiated per second by the body if the body temperature is 37°C. Stefan constant σ is $6.0 \times 10^{-8}\ W\,m^{-2}\,K^{-4}$.

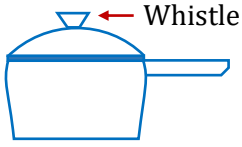
PEH214

KTG

26. An insulated container containing monoatomic gas of molar mass m is moving with a velocity v_0. If the container is suddenly stopped, find the change in temperature.

PEH265

27. An empty pressure cooker of volume 10 litres contains air at atmospheric pressure 10^5 Pa and temperature of 27°C. It contains a whistle which has area of 0.1 cm^2 and weight of 100 g. What should be the temperature of air inside so that the whistle is just lifted up?



PEH267

28. Find the molecular mass of a gas if the specific heats of the gas are C_P=0.2 $cal/gm°C$ and C_V = 0.15 $cal/gm°C$. [Take R = 2 cal/mole°C]

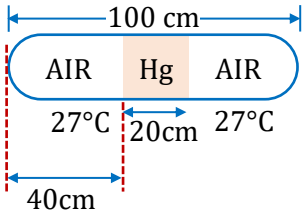
PEH268

For More Join: @IITJEE_Advanced

29. 16 g of oxygen at 37°C is mixed with 14 g of nitrogen at 27°C. Find the temperature of the mixture?

PEH269

30. In the given figure a glass tube lies horizontally with the middle 20 cm containing mercury. The two ends of the tube contains air at 27°C and at a pressure 76 cm of mercury. Now the air column on one side is maintained at 0°C and the other side is maintained at 127°C. Find the new length of the air column on the cooler side. Neglect the changes in the volume of mercury and of the glass.



PEH270

31. Two vessels A and B, thermally insulated, contain an ideal monoatomic gas. A small tube fitted with a valve connects these vessels. Initially the vessel A has 2 liters of gas at 300 K and 2×10^5 $N\,m^{-2}$ pressure while vessel B has 4 liters of gas at 350 K and 4×10^5 Nm^{-2} pressure. The valve is now opened and the system reaches equilibrium in pressure and temperature. Calculate the new pressure and temperature. ($R = \frac{25}{3}$ $J/mol\text{-}K$)

PEH271

32. Find the maximum attainable temperature of an ideal gas in the following process : where (a) $p = p_0 - \alpha V^2$; (b) $p = p_0 e^{-\beta V}$, where p_0, α and β are positive constants and V is the volume of one mole of gas.

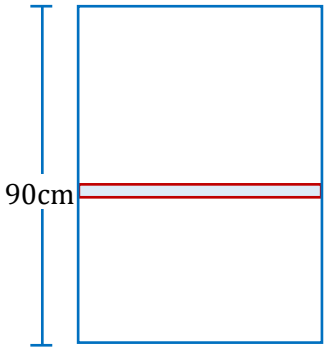
Pankers Academy JEE

PEH272

33. An ideal gas of molar mass M is contained in a tall vertical cylindrical vessel whose base area is S and height h. The temperature of the gas is T, its pressure on the bottom base is p_0. Assuming the temperature and the free-fall acceleration g to be independent of the height, find the mass of gas in the vessel.

PEH273

34. Figure shows a vertical cylindrical vessel separated in two parts by a frictionless piston free to move along the length of the vessel. The length of the cylinder is 90 cm and the piston divides the cylinder in the ratio of 5 : 4. Each of the two parts of the vessel contains 0.1 mole of an ideal gas. The temperature of the gas is 300 K in each part. Calculate the mass of the piston.

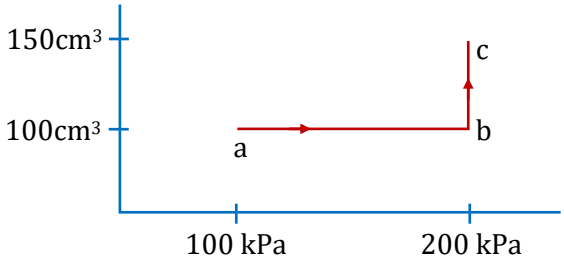


PEH400

For More Join: @IITJEE_Advanced

Work Done by Gas and First Law of Thermodynamics

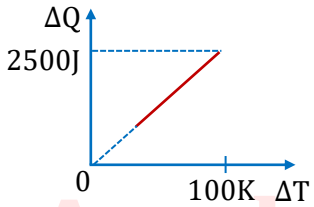
35. An ideal gas ($\gamma = 1.67$) is taken through the process abc shown in figure. The temperature at the point a is 300 *K*. Calculate



(a) the temperature at *b* and *c*,
(b) the work done in the process,
(c) the amount of heat supplied in the path *ab* and in the path *bc*
(d) the change in the internal energy of the gas in the process.

PEH274

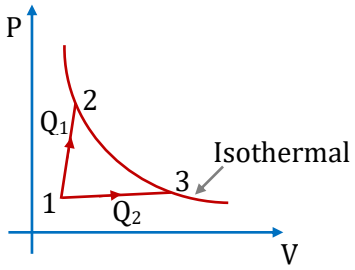
36. One mole of a gas mixture is heated under constant pressure, and heat required ΔQ is plotted against temperature difference acquired. Find the value of γ for mixture.



PEH275

Rankers Academy JEE

37. A gas takes part in two processes in which it is heated from the same initial state 1 to the same final temperature. The processes are shown on the *P-V* diagram by the straight line 1-2 and 1-3. 2 and 3 are the points on the same isothermal curve. Q_1 and Q_2 are the heat transfer along the two processes. Then in which case will the heat transfer be more.



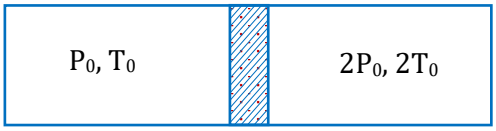
PEH276

38. Two containers *A* and *B* of equal volume V_0 / 2 each are connected by a narrow tube which can be closed by a valve. The containers are fitted with pistons which can be moved to change the volumes. Initially, the valve is open and the containers contain an ideal gas ($C_p/C_v = \gamma$) at atmospheric pressure P_0 and atmospheric temperature $2T_0$. The walls of the containers *A* are highly conducting and of *B* are non-conducting. The valve is now closed and the pistons are slowly pulled out to increase the volumes of the containers to double the original value. (a) Calculate the temperatures and pressures in the two containers. (b) The valve is now opened for sufficient time so that the gases acquire a common temperature and pressure. Find the new values of the temperature and the pressure.

PEH277

For More Join: @IITJEE_Advanced

39. A cylindrical tube with adiabatic walls having volume $2V_0$ contains an ideal monoatomic gas as shown in figure. The tube is divided into two equal parts by a fixed super conducting wall. Initially, the pressure and the temperature are P_0, T_0 on the left and $2P_0$, $2T_0$ on the right. When system is left for sufficient amount of time the temperature on both sides becomes equal

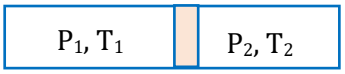


(a) Find work done by the gas on the right part?
(b) Find the final pressures on the two sides.
(c) Find the final equilibrium temperature.
(d) How much heat has flown from the gas on the right to the gas on the left?

PEH278

40. In given figure, an adiabatic cylindrical tube of volume $2V_0$ is divided in two equal parts by a frictionless adiabatic separator. An ideal gas in left side of a tube having pressure P_1 and temperature T_1 where as in the right side having pressure P_2 and temperature T_2. $C_p/C_v = \gamma$ is the same for both the gases. The separator is slid slowly and is released at a position where it can stay in equilibrium. Find;

(a) the final volumes of the two parts,
(b) the heat given to the gas in the left part and
(c) the final common pressure of the gases.



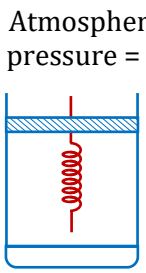
PEH279

Rankers Academy JEE

41. An amount n (in moles) of a monatomic gas at an initial temperature T_0 is enclosed in a cylindrical vessel fitted with a light piston. The surrounding air has a temperature $T_s(> T_0)$ and the atmospheric pressure is pa. Heat may be conducted between the surrounding and the gas through the bottom of the cylinder. The bottom has a surface area A, thickness x and thermal conductivity K. Assuming all changes to be slow, find the distance moved by the piston in time t.

PEH280

42. Consider one mole of perfect gas in a cylinder of unit cross section with a piston attached (Fig.). A spring (spring constant k) is attached (unstretched length L) to the piston and to the bottom of the cylinder. Initially the spring is unstretched and the gas is in equilibrium. A certain amount of heat Q is supplied to the gas causing an increase of volume from V_0 to V_1.

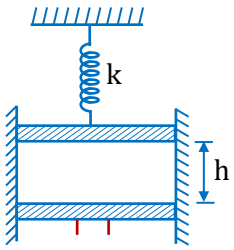


(a) What is the initial pressure of the system?
(b) What is the final pressure of the system?
(c) Using the first law of thermodynamics, write down a relation between Q, P_a, V, V_0 and k.

PEH281

For More Join: @IITJEE_Advanced

43. An ideal gas at *NTP* is enclosed in a adiabatic vertical cylinder having area of cross section $A = 27cm^2$, between two light movable pistons as shown in the figure. Spring with force constant $k = 3700\ N/m$ is in a relaxed state initially. Now the lower piston is moved upwards a height $h/2$, h being the initial length of gas column. It is observed that the upper piston moves up by a distance $h/16$. Find h taking γ for the gas to be 1.5. Also find the final temperature of the gas.

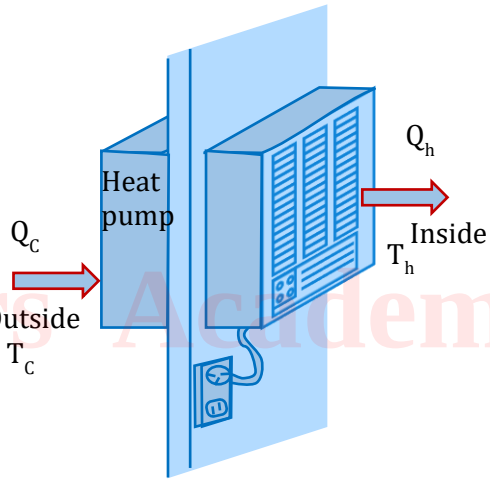


PEH282

Second Law of Thermodynamics

44. A heat pump used for heating shown in figure is essentially an air conditioner installed backward. It extracts energy from colder air outside and deposits it in a warmer room. Suppose the ratio of the actual energy entering the room to the work done by the device's motor is 20.0% of the theoretical maximum ratio. Determine the energy entering the room per joule of work done by the motor, given that the inside temperature is 27.0°C and the outside temperature is –3°C.

PEH283



Rankers Academy JEE

45. A Carnot engine working between 400 *K* and 800 *K* has a work output of 1200 *J* per cycle. What is the amount of heat energy supplied to the engine from source per cycle?

PEH284

46. The temperature insides and outside of refrigerator are 260 *K* and 315 *K* respectively. Assuming that the refrigerator cycle is reversible, calculate the heat delivered to surroundings for every joule of work done.

PEH285

Entropy

47. Find the entropy increment of one mole of carbon dioxide when its absolute temperature increases $n = 2.0$ times if the process of heating is : ($C_v = 27\ J/mol\text{-}K$)

(a) isochoric ; (b) isobaric.

The gas is to be regarded as ideal.

PEH286

48. A quantity of 0.20 mol of argon gas contained in a volume of 5.0 *L* mixes with 0.50 mol of neon gas contained in a volume of 5*L*, making a total volume of 10 *L*. Both gases are at the same temperature and can be regarded as ideal. Each volume is thermally isolated. What is the change in entropy ?

PEH287

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. A copper ball of mass 100 *gm* is at a temperature *T*. It is dropped in a copper calorimeter of mass 100*gm*, filled with 170 *gm* of water at room temperature. Subsequently, the temperature of the system is found to be 75°*C*. *T* is given by :

(Given : room temperature = 30° *C*, specific heat of copper = 0.1 *cal/gm°C*) **[JEE (Main) 2017]**

(1) 1250°*C* (2) 825°*C* (3) 800°*C* (4) 885° *C*

PEH022

2. C_p and C_v are specific heats at constant pressure and constant volume respectively. It is observed that

$C_p - C_v = a$ for hydrogen gas

$C_p - C_v = b$ for nitrogen gas

The correct relation between *a* and *b* is : **[JEE (Main) 2017]**

(1) $a = 14\,b$ (2) $a = 28\,b$ (3) $a = \frac{1}{14}\,b$ (4) $a = b$

PEH290

3. Two moles of an ideal monoatomic gas occupies a volume *V* at 27°*C*. The gas expands adiabatically to a volume 2*V*. Calculate (a) the final temperature of the gas and (b) change in its internal energy. **[JEE (Main) 2018]**

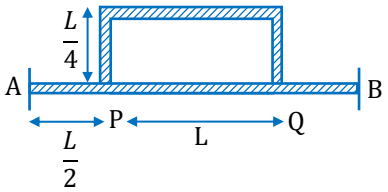
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(1) (a) 195 *K* (b) –2.7 *kJ* (2) (a) 189 *K* (b) –2.7 *kJ*

(3) (a) 195 *K* (b) 2.7 *kJ* (4) (a) 189 *K* (b) 2.7 *kJ*

PEH291

4. Temperature difference of 120°*C* is maintained between two ends of a uniform rod *AB* of length 2*L*. Another bent rod *PQ*, of same cross-section as *AB* and length, is connected across *AB* (See figure). In steady state, temperature difference between *P* and *Q* will be close to : **[JEE (Main) 2019]**



(1) 60°*C* (2) 75°*C* (3) 35°*C* (4) 45°*C*

PEH023

5. A thermometer graduated according to a linear scale reads a value x_0 when in contact with boiling water, and $x_0/3$ when in contact with ice. What is the temperature of an object in °*C*, if this thermometer in the contact with the object reads $x_0/2$? **[JEE (Main) 2019]**

(1) 35 (2) 25 (3) 60 (4) 40

PEH024

For More Join: @IITJEE_Advanced

6. A cylinder of radius R is surrounded by a cylindrical shell of inner radius R and outer radius $2R$. The thermal conductivity of the material of the inner cylinder is K_1 and that of the outer cylinder is K_2. Assuming no loss of heat, the effective thermal conductivity of the system for heat flowing along the length of the cylinder is: **[JEE (Main) 2019]**

(1) $K_1 + K_2$ (2) $\frac{K_1 + K_2}{2}$ (3) $\frac{2K_1 + 3K_2}{5}$ (4) $\frac{K_1 + 3K_2}{4}$

PEH025

7. A bakelite beaker has volume capacity of 500 cc at 30°C. When it is partially filled with V_m volume (at 30°) of mercury, it is found that the unfilled volume of the beaker remains constant as temperature is varied. If $\gamma_{(beaker)} = 6 \times 10^{-6}\,°C^{-1}$ and $\gamma_{(mercury)} = 1.5 \times 10^{-4}\,°C^{-1}$, where γ is the coefficient of volume expansion, then V_m (in cc) is close to_____. **[JEE (Main) 2020]**

PEH027

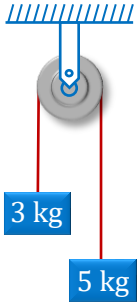
8. Under an adiabatic process, the volume of an ideal gas gets doubled. Consequently the mean collision time between the gas molecule changes from τ_1 to τ_2. If $\frac{C_p}{C_v} = \gamma$ for this gas then a good estimate for $\frac{\tau_2}{\tau_1}$ is given by : **[JEE (Main) 2020]**

(1) $\left(\frac{1}{2}\right)^{\frac{\gamma+1}{2}}$ (2) 2 (3) $\frac{1}{2}$ (4) $\left(\frac{1}{2}\right)^{\gamma}$

PEH292

9. Two blocks of masses 3 kg and 5 kg are connected by a metal wire going over a smooth pulley. The breaking stress of the metal is $\frac{24}{\pi} \times 10^2\ Nm^{-2}$. What is the minimum radius of the wire?

(Take $g = 10\ ms^{-2}$) **[JEE (Main) 2021]**



(1) 125 cm (2) 1250 cm (3) 12.5 cm (4) 1.25 cm

PEH028

10. In 5 minutes, a body cools from 75°C to 65°C at room temperature of 25°C. The temperature of body at the end of next 5 minutes is _____°C. **[JEE (Main) 2021]**

PEH029

11. A monoatomic gas performs a work of $\frac{Q}{4}$ where Q is the heat supplied to it. The molar heat capacity of the gas will be _____R during this transformation. **[JEE (Main) 2022]**

Where R is the gas constant.

PEH293

For More Join: @IITJEE_Advanced

12. The elastic potential energy stored in a steel wire of length 20 m stretched through 2 cm is 80 J. The cross sectional area of the wire is_____ mm^2.

(Given, $y = 2.0 \times 10^{11} Nm^{-2}$) **[JEE (Main) 2023]**

PEH173

13. An air bubble of volume 1 cm^3 rises from the bottom of a lake 40m deep to the surface at a temperature of 12°C. The atmospheric pressure is 1×10^5 Pa, the density of water is 1000 kg/m^3 and g = 10 m/s^2. There is no difference of the temperature of water at the depth of 40 m and on the surface. The volume of air bubble when it reaches the surface will be **[JEE (Main) 2023]**

(1) 5 cm^3 (2) 2 cm^3 (3) 4 cm^3 (4) 3 cm^3

PEH294

14. An elastic spring under tension of 3 N has a length a. Its length is b under tension 2 N. For its length $(3a - 2b)$, the value of tension will be _____ N. **[JEE (Main) 2024]**

PEH215

15. On celcius scale the temperature of body increases by 40°C. The increase in temperature on Fahrenheit scale is: **[JEE (Main) 2024]**

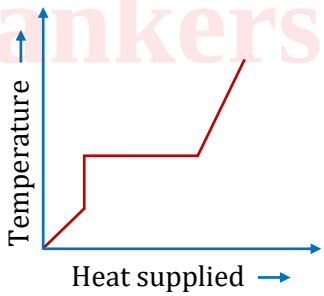
(1) 70°F (2) 68°F (3) 72°F (4) 75°F

PEH216

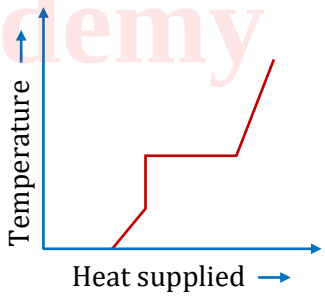
16. A block of ice at –10°C is slowly heated and converted to steam at 100°C. Which of the following curves represent the phenomenon qualitatively: **[JEE (Main) 2024]**

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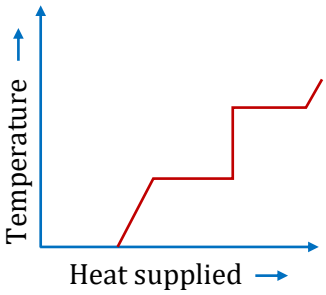
(1)



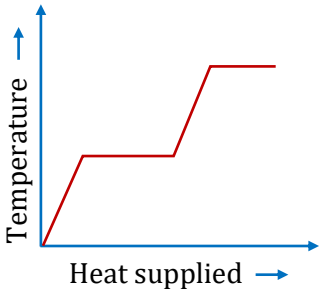
(2)



(3)



(4)



PEH217

17. If the root mean square velocity of hydrogen molecule at a given temperature and pressure is 2 km/s, the root mean square velocity of oxygen at the same condition in km/s is :

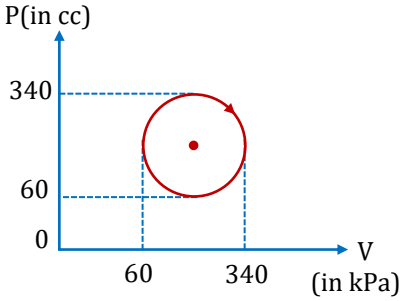
[JEE (Main) 2024]

(1) 2.0 (2) 0.5 (3) 1.5 (4) 1.0

PEH218

For More Join: @IITJEE_Advanced

18. The heat absorbed by a system in going through the given cyclic process is :

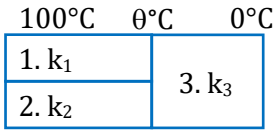


[JEE (Main) 2024]

(1) 61.6 J (2) 431.2 J (3) 616 J (4) 19.6 J

PEH219

19. Three conductions of same length having thermal conductivity k_1, k_2 and k_3 are connected as shown in figure.



Area of cross sections of 1st and 2nd conductor are same and for 3rd conductor it is double of the 1st conductor. The temperatures are given in the figure. In steady state condition, the value of θ is _____ °C.

(Given : $k_1 = 60\ Js^{-1}m^{-1}K^{-1}$, $k_2 = 120\ Js^{-1}m^{-1}K^{-1}$, $k_3 = 135\ Js^{-1}m^{-1}K^{-1}$) **[JEE (Main) 2025]**

PEH393

20. Water of mass m gram is slowly heated to increase the temperature from T_1 to T_2. The change in entropy of the water, given specific heat of water is 1 $Jkg^{-1}K^{-1}$, is: **[JEE (Main) 2025]**

(1) zero (2) $m\,(T_2 - T_1)$ (3) $m\ln\left(\frac{T_1}{T_2}\right)$ (4) $m\ln\left(\frac{T_2}{T_1}\right)$

PEH394

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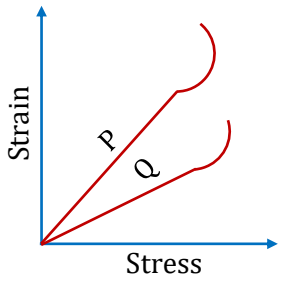
For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. Two spherical stars A and B emit blackbody radiation. The radius of A is 400 times that of B and A emits 10^4 times the power emitted from B. The ratio $\left(\frac{\lambda_A}{\lambda_B}\right)$ of their wavelengths λ_A and λ_B at which the peaks occur in their respective radiation curves is. **[JEE (Advanced) 2015]**

PEH045

2. In plotting stress versus strain curves for two materials P and Q, a student by mistake puts strain on the y-axis and stress on the x-axis as shown in the figure. Then the correct statement(s) is (are): **[JEE (Advanced) 2015]**

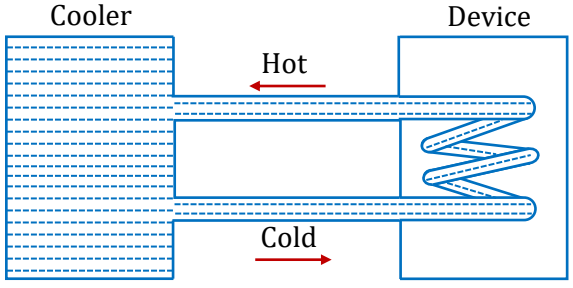


(A) P has more tensile strength than Q
(B) P is more ductile than Q
(C) P is more brittle than Q
(D) The Young's modulus of P is more than that of Q

PEH046

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3. A water cooler of storage capacity 120 litres can cool water at constant rate of P watts. In a closed circulation system (as shown schematically in the figure), the water from the cooler is used to cool an external device that generates constantly 3 kW of heat (thermal load). The temperature of water fed into the device cannot exceed 30°C and the entire stored 120 litres of water is initially cooled to 10°C. The entire system is thermally insulated. The minimum value of P (in watts) for which the device can be operated for 3 hours is : **[JEE (Advanced) 2016]**



(Specific heat of water is 4.2 $kJ\ kg^{-1}\ K^{-1}$ and the density of water is 1000 $kg\ m^{-3}$)
(A) 1600 (B) 2067 (C) 2533 (D) 3933

PEH047

4. A metal is heated in a furnace where a sensor is kept above the metal surface to read the power radiated (P) by the metal. The sensor has a scale that displays $\log_2(P/P_0)$, where P_0 is a constant. When the metal surface is at a temperature of 487°C, the sensor shows a value 1. Assume that the emissivity of the metallic surface remains constant. What is the value displayed by the sensor when the temperature of the metal surface is raised to 2767°C ? **[JEE (Advanced) 2016]**

PEH048

For More Join: @IITJEE_Advanced

5. The ends Q and R of two thin wires, PQ and RS, are soldered (joined) together. Initially each of the wires has a length of $1m$ at $10°C$. Now the end P is maintained at $10°C$, while the end S is heated and maintained at $400°C$. The system is thermally insulated from its surroundings. If the thermal conductivity of wire PQ is twice that of the wire RS and the coefficient of linear thermal expansion of PQ is $1.2 \times 10^{-5} K^{-1}$, the change in length of the wire PQ is **[JEE (Advanced) 2016]**

(A) 0.78 mm (B) 0.90 mm (C) 1.56 mm (D) 2.34 mm

PEH049

6. A gas is enclosed in a cylinder with a movable frictionless piston. Its initial thermodynamic state at pressure $P_i = 10^5\ Pa$ and volume $V_i = 10^{-3}\ m^3$ changes to a final state at $P_f = \left(\frac{1}{32}\right) \times 10^5 Pa$ and $V_f = 8 \times 10^{-3}\ m^3$ in an adiabatic quasi-static process, such that P^3V^5 = constant. Consider another thermodynamic process that brings the system from the same initial state to the same final state in two steps: an isobaric expansion at P_i followed by an isochoric (isovolumetric) process at volumes V_f. The amount of heat supplied to the system in the two step process is approximately

[JEE (Advanced) 2016]

(A) 112 J (B) 294 J (C) 588 J (D) 813 J

PEH295

7. A human body has a surface area of approximately 1 m^2. The normal body temperature is 10 K above the surrounding room temperature T_0. Take the room temperature to be $T_0 = 300\ K$. For $T_0 = 300\ K$, the value of $\sigma T_0^4 = 460\ Wm^{-2}$ (where σ is the Stefan-Boltzmann constant). Which of the following options is/are **correct** ? **[JEE (Advanced) 2017]**

(A) The amount of energy radiated by the body in 1 second is close to 60 Joules

(B) If the surrounding temperature reduces by a small amount $\Delta T_0 << T_0$, then to maintain the same body temperature the same (living) human being needs to radiate ΔW = more energy per unit time

(C) Reducing the exposed surface area of the body (e.g. by curling up) allows humans to maintain the same body temperature while reducing the energy lost by radiation

(D) If the body temperature rises significantly then the peak in the spectrum of electromagnetic radiation emitted by the body would shift to longer wavelengths

PEH050

8. A flat plate is moving normal to its plane through a gas under the action of a constant force F. The gas is kept at a very low pressure. The speed of the plate v is much less than the average speed u of the gas molecules. Which of the following options is/are true ? **[JEE (Advanced) 2017]**

(A) The resistive force experienced by the plate is proportional to v

(B) The pressure difference between the leading and trailing faces of the plate is proportional to uv.

(C) The plate will continue to move with constant non-zero acceleration, at all times

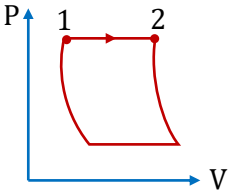
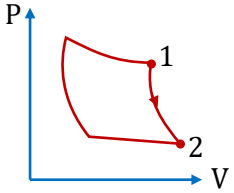
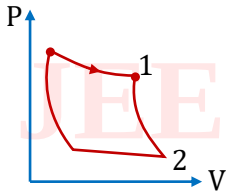
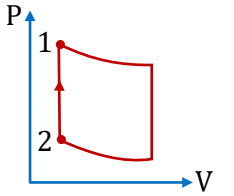
(D) At a later time the external force F balances the resistive force.

PEH296

For More Join: @IITJEE_Advanced

Answer Q.9, Q.10 and Q.11 by appropriately matching the information given in the three columns of the following table.

An ideal gas is undergoing a cyclic thermodynamics process in different ways as shown in the corresponding P–V diagrams in column 3 of the table. Consider only the path from state 1 to state 2. W denotes the corresponding work done on the system. The equations and plots in the table have standard notations as used in thermodynamics processes. Here γ is the ratio of heat capacities at constant pressure and constant volume. The number of moles in the gas is n. **[JEE (Advanced) 2017]**

	Column-1		**Column-2**		**Column-3**
(I)	$W_{1\to2}=\frac{1}{\gamma-1}(P_2V_2-P_1V_1)$	(i)	Isothermal	(P)	
(II)	$W_{1\to2}=-PV_2+PV_1$	(ii)	Isochoric	(Q)	
(III)	$W_{1\to2}=0$	(iii)	Isobaric	(R)	
(IV)	$W_{1\to2}=-nRT\ln\frac{V_2}{V_1}$	(iv)	Adiabatic	(S)	

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9. Which of the following options is the only correct representation of a process in which $\Delta U=\Delta Q-P\Delta V$?

(A) (II) (iv) (R) (B) (II) (iii) (P) (C) (II) (iii) (S) (D) (III) (iii) (P)

PEH297

10. Which one of the following options is the correct combination ?

(A) (III) (ii) (S) (B) (II) (iv) (R) (C) (II) (iv) (P) (D) (IV) (ii) (S)

PEH298

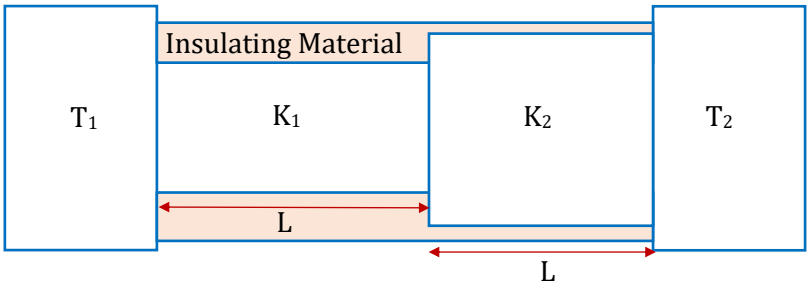
11. Which one of the following options correctly represents a thermodynamics process that is used as a correction in the determination of the speed of sound in an ideal gas ?

(A) (III) (iv) (R) (B) (I) (ii) (Q) (C) (IV) (ii) (R) (D) (I) (iv) (Q)

PEH299

For More Join: @IITJEE_Advanced

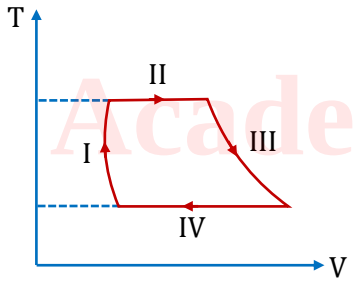
12. Two conducting cylinders of equal length but different radii are connected in series between two heat baths kept at temperatures $T_1 = 300\ K$ and $T_2 = 100\ K$, as shown in the figure. The radius of the bigger cylinder is twice that of the smaller one and the thermal conductivities of the materials of the smaller and the larger cylinders are K_1 and K_2 respectively. If the temperature at the junction of the two cylinders in the steady state is 200 K, then K_1/K_2 = _____. **[JEE (Advanced) 2018]**



PEH051

13. One mole of a monatomic ideal gas undergoes a cyclic process as shown in the figure (where V is the volume and T is the temperature). Which of the statements below is (are) true ?

[JEE (Advanced) 2018]



(A) Process I is an isochoric process

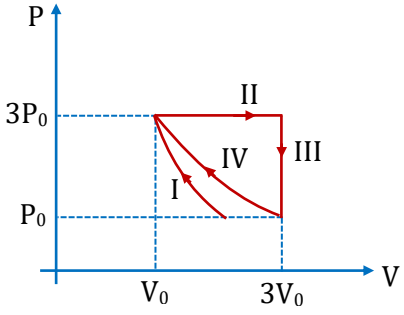
(B) In process II, gas absorbs heat

(C) In process IV, gas releases heat

(D) Processes I and II are not isobaric

PEH300

14. One mole of a monatomic ideal gas undergoes four thermodynamic processes as shown schematically in the PV-diagram below. Among these four processes, one is isobaric, one is isochoric, one is isothermal and one is adiabatic. Match the processes mentioned in List-I with the corresponding statements in List–II. **[JEE (Advanced) 2018]**



For More Join: @IITJEE_Advanced

List–I	List–II
P. In process I	1. Work done by the gas is zero
Q. In process II	2. Temperature of the gas remains unchanged
R. In process III	3. No heat is exchanged between the gas and its surroundings
S. In process IV	4. Work done by the gas is $6\,P_0V_0$

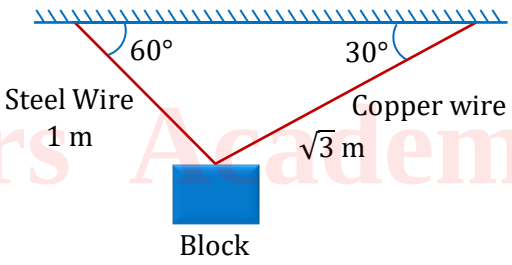
(A) P → 4 ; Q → 3 ; R → 1 ; S → 2
(B) P → 1 ; Q → 3 ; R → 2 ; S → 4
(C) P → 3 ; Q → 4 ; R → 1 ; S → 2
(D) P → 3 ; Q → 4 ; R → 2 ; S → 1

PEH301

15. A block of weight 100 N is suspended by copper and steel wires of same cross sectional area 0.5 cm^2 and, length $\sqrt{3}\,m$ and 1 m, respectively. Their other ends are fixed on a ceiling as shown in figure. The angles subtended by copper and steel wires with ceiling are 30° and 60°, respectively. If elongation in copper wire is $(\Delta\ell_C)$ and elongation in steel wire is $(\Delta\ell_S)$, then the ratio $\frac{\Delta\ell_C}{\Delta\ell_S}$ is ____.

[Young's modulus for copper and steel are $1 \times 10^{11}\ N/m^2$ and $2 \times 10^{11}\ N/m^2$ respectively]

[JEE (Advanced) 2019]



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PEH052

16. A liquid at 30°C is poured very slowly into a Calorimeter that is at temperature of 110°C. The boiling temperature of the liquid is 80°C. It is found that the first 5 gm of the liquid completely evaporates. After pouring another 80 gm of the liquid the equilibrium temperature is found to be 50°C. The ratio of the Latent heat of the liquid to its specific heat will be _____°C.

[Neglect the heat exchange with surrounding] **[JEE (Advanced) 2019]**

PEH053

17. A mixture of ideal gas containing 5 moles of monatomic gas and 1 mole of rigid diatomic gas is initially at pressure P_0, volume V_0 and temperature T_0. If the gas mixture is adiabatically compressed to a volume $V_0/4$, then the correct statement(s) is/are,

(Give $2^{1.2} = 2.3$; $2^{3.2} = 9.2$; R is gas constant) **[JEE (Advanced) 2019]**

(A) The final pressure of the gas mixture after compression is in between $9P_0$ and $10P_0$

(B) The average kinetic energy of the gas mixture after compression is in between $18RT_0$ and $19RT_0$

(C) The work $|W|$ done during the process is $13RT_0$

(D) Adiabatic constant of the gas mixture is 1.6

PEH302

For More Join: @IITJEE_Advanced

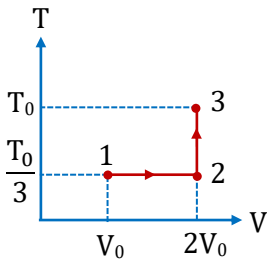
18. **Answer the following by appropriately matching the lists based on the information given in the paragraph.** **[JEE (Advanced) 2019]**

In a thermodynamic process on an ideal monatomic gas, the infinitesimal heat absorbed by the gas is given by $T\Delta X$, where T is temperature of the system and ΔX is the infinitesimal change in a thermodynamic quantity X of the system. For a mole of monatomic ideal gas $X = \frac{3}{2}R\ln\left(\frac{T}{T_A}\right)+\left(\frac{V}{V_A}\right)R\ln$. Here, R is gas constant, V is volume of gas, T_A and V_A are constants.

	List-I		**List-II**
(I)	Work done by the system in process $1 \to 2 \to 3$	(P)	$\frac{1}{3}RT_0\ln 2$
(II)	Change in internal energy in process $1 \to 2 \to 3$	(Q)	$\frac{1}{3}RT_0$
(III)	Heat absorbed by the system in process $1 \to 2 \to 3$	(R)	RT_0
(IV)	Heat absorbed by the system in process $1 \to 2$	(S)	$\frac{4}{3}RT_0$
		(T)	$\frac{1}{3}RT_0(3+\ln 2)$
		(U)	$\frac{5}{6}RT_0$

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If the process on one mole of monatomic ideal gas is an shown is as shown in the TV-diagram with $P_0V_0 = \frac{1}{3}RT_0$, the correct match is



(A) I $\to$ S, II $\to$ T, III $\to$ Q, IV $\to$ U

(B) I $\to$ P, II $\to$ R, III $\to$ T, IV $\to$ S

(C) I $\to$ P, II $\to$ T, III $\to$ Q, IV $\to$ T

(D) I $\to$ P, II $\to$ R, III $\to$ T, IV $\to$ P

PEH303

19. Consider one mole of helium gas enclosed in a container at initial pressure P_1 and volume V_1. It expands isothermally to volume $4V_1$. After this, the gas expands adiabatically and its volume becomes $32V_1$. The work done by the gas during isothermal and adiabatic expansion processes are W_{iso} and W_{adia}, respectively. If the ratio $\frac{W_{iso}}{W_{adia}} = f\ln 2$, then f is _____. **[JEE (Advanced) 2020]**

PEH304

For More Join: @IITJEE_Advanced

20. The filament of a light bulb has surface area 64 mm^2. The filament can be considered as a black body at temperature 2500 K emitting radiation like a point source when viewed from far. At night the light bulb is observed from a distance of 100 m. Assume the pupil of the eyes of the observer to be circular with radius 3 mm. Then

(Take Stefan-Boltzmann constant = $5.67 \times 10^{-8}\ Wm^{-2}\ K^{-4}$, Wien's displacement constant = $2.90 \times 10^{-3}\ m\text{-}K$, Planck's constant = $6.63 \times 10^{-34}\ Js$, speed of light in vacuum = $3.00 \times 10^{8}\ ms^{-1}$)-

[JEE (Advanced) 2021]

(A) power radiated by the filament is in the range 642 W to 645 W

(B) radiated power entering into one eye of the observer is in the range $3.15 \times 10^{-8}\ W$ to $3.25 \times 10^{-8}\ W$

(C) the wavelength corresponding to the maximum intensity of light is 1160 nm

(D) taking the average wavelength of emitted radiation to be 1740 nm, the total number of photons entering per second into one eye of the observer is in the range 2.75×10^{11} to 2.85×10^{11}

PEH054

21. A container with 1 kg of water in it is kept in sunlight, which causes the water to get warmer than the surroundings. The average energy per unit time per unit area received due to the sunlight is $700\ Wm^{-2}$ and it is absorbed by the water over an effective area of 0.05 m^2. Assuming that the heat loss from the water to the surroundings is governed by Newton's law of cooling, the difference (in °C) in the temperature of water and the surroundings after a long time will be _____.

(Ignore effect of the container, and take constant for Newton's law of cooling = 0.001 s^{-1}, Heat capacity of water = 4200 $J\ kg^{-1}\ K^{-1}$) **[JEE (Advanced) 2021]**

PEH055

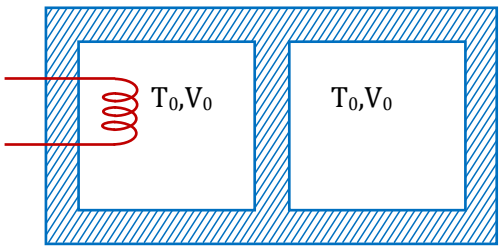
Rankers Academy JEE

22. A small object is placed at the center of a large evacuated hollow spherical container. Assume that the container is maintained at 0 K. At time $t = 0$, the temperature of the object is 200 K. The temperature of the object becomes 100 K at $t = t_1$ and 50 K at $t = t_2$. Assume the object and the container to be ideal black bodies. The heat capacity of the object does not depend on temperature. The ratio (t_2/t_1) is_____. **[JEE (Advanced) 2021]**

PEH056

PARAGRAPH FOR QUESTION NO. 23 and 24

A thermally insulating cylinder has a thermally insulating and frictionless movable partition in the middle, as shown in the figure below. On each side of the partition, there is one mole of an ideal gas, with specific heat at constant volume, $C_v = 2R$. Here, R is the gas constant. Initially, each side has a volume V_0 and temperature T_0. The left side has an electric heater, which is turned on at very low power to transfer heat Q to the gas on the left side. As a result the partition moves slowly towards the right reducing the right side volume to $V_0/2$. Consequently, the gas temperatures on the left and the right sides become T_L and T_R, respectively. Ignore the changes in the temperatures of the cylinder, heater and the partition. **[JEE (Advanced) 2021]**



For More Join: @IITJEE_Advanced

23. The value of $\frac{T_R}{T_0}$ is

(A) $\sqrt{2}$ (B) $\sqrt{3}$ (C) 2 (D) 3

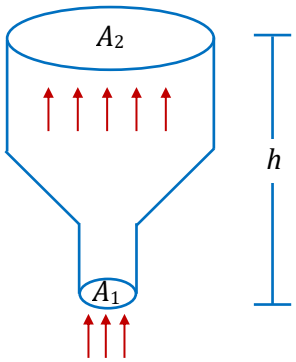
PEH305

24. The value of $\frac{Q}{RT_0}$ is

(A) $4\left(2\sqrt{2}+1\right)$ (B) $4\left(2\sqrt{2}-1\right)$ (C) $\left(5\sqrt{2}+1\right)$ (D) $\left(5\sqrt{2}-1\right)$

PEH306

25. An ideal gas of density $\rho = 0.2\ kg\ m^{-3}$ enters a chimney of height h at the rate of $\alpha = 0.8 kgs^{-1}$ from its lower end, and escapes through the upper end as shown in the figure. The cross-sectional area of the lower end is $A_1 = 0.1\ m^2$ and the upper end is $A_2 = 0.4\ m^2$. The pressure and the temperature of the gas at the lower end are 600 *Pa* and 300 *K*, respectively, while its temperature at the upper end is 150 *K*. The chimney is heat insulated so that the gas undergoes adiabatic expansion. Take $g = 10\ ms^{-2}$ and the ratio of specific heats of the gas $\gamma = 2$. Ignore atmospheric pressure. **[JEE (Advanced) 2022]**



Which of the following statement(s) is(are) correct?

(A) The pressure of the gas at the upper end of the chimney is 300 *Pa*.

(B) The velocity of the gas at the lower end of the chimney is 40 ms^{-1} and at the upper end is 20 ms^{-1}.

(C) The height of the chimney is 590 *m*.

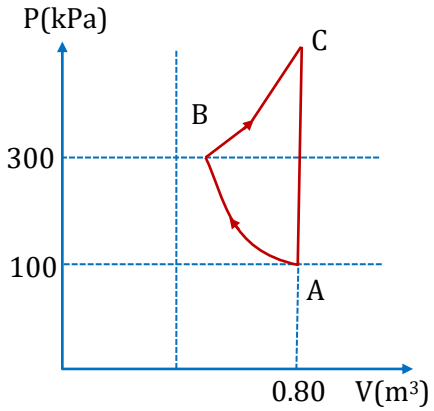
(D) The density of the gas at the upper end is 0.05 $kg\ m^{-3}$.

PEH307

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26. In the given *P-V* diagram, a monoatomic gas $\left(\gamma = \frac{5}{3}\right)$ is first compressed adiabatically from state *A* to state *B*. Then it expands isothermally from state *B* to state *C*. [Given: $\left(\frac{1}{3}\right)^{0.6} \simeq 0.5, \ln 2 \simeq 0.7$].

[JEE (Advanced) 2022]



Which of the following statement(s) is(are) correct?

(A) The magnitude of the total work done in the process $A \to B \to C$ is 144 *kJ*.

(B) The magnitude of the work done in the process $B \to C$ is 84 *kJ*.

(C) The magnitude of the work done in the process $A \to B$ is 60 *kJ*.

(D) The magnitude of the work done in the process $C \to A$ is zero.

PEH308

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27. Young's modulus of elasticity Y is expressed in terms of three derived quantities, namely, the gravitational constant G, Planck's constant h and the speed of light c, as $Y = c^{\alpha} h^{\beta} G^{\gamma}$. Which of the following is the correct option ? **[JEE (Advanced) 2023]**

(A) $\alpha = 7, \beta = -1, \gamma = -2$ (B) $\alpha = -7, \beta = -1, \gamma = -2$

(C) $\alpha = 7, \beta = -1, \gamma = 2$ (D) $\alpha = -7, \beta = 1, \gamma = -2$

PEH174

28. One mole of an ideal gas expands adiabatically from an initial state (T_A, V_0) to final state (T_f, $5V_0$). Another mole of the same gas expands isothermally from a different initial state (T_B, V_0) to the same final state (T_f, $5V_0$). The ratio of the specific heats at constant pressure and constant volume of this ideal gas is γ. What is the ratio T_A/T_B? **[JEE (Advanced) 2023]**

(A) $5^{\gamma-1}$ (B) $5^{1-\gamma}$ (C) 5^{γ} (D) $5^{1+\gamma}$

PEH309

29. A closed container contains a homogeneous mixture of two moles of an ideal monatomic gas ($\gamma = 5/3$) and one mole of an ideal diatomic gas ($\gamma = 7/5$). Here, γ is the ratio of the specific heats at constant pressure and constant volume of an ideal gas. The gas mixture does a work of 66 Joule when heated at constant pressure. The change in its internal energy is _____ Joule. **[JEE (Advanced) 2023]**

PEH310

30. An ideal gas is in thermodynamic equilibrium. The number of degrees of freedom of a molecule of the gas in n. The internal energy of one mole of the gas is U_n and the speed of sound in the gas is v_n. At a fixed temperature and pressure, which of the following is the correct option ? **[JEE (Advanced) 2023]**

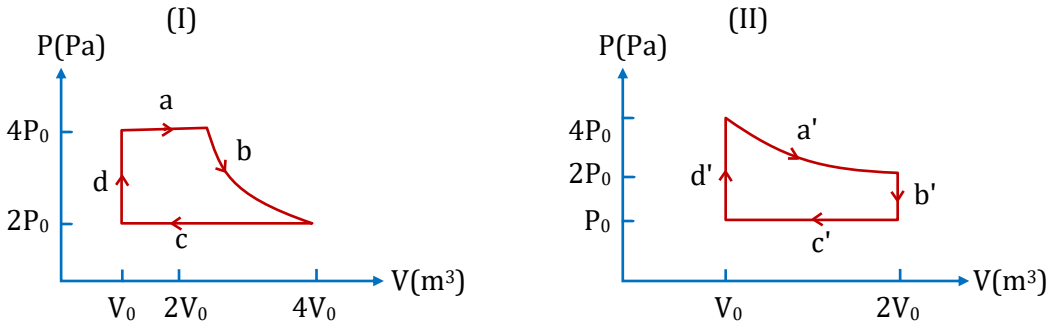
(A) $v_3 < v_6$ and $U_3 > U_6$ (B) $v_5 > v_3$ and $U_3 > U_5$

(C) $v_5 > v_7$ and $U_5 < U_7$ (D) $v_6 < v_7$ and $U_6 < U_7$

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PEH311

31. One mole of an ideal gas undergoes two different cyclic processes I and II, as shown in the P-V diagrams below. In cycle I, processes a, b, c and d are isobaric, isothermal, isobaric and isochoric, respectively. In cycle II, processes a', b', c' and d' are isothermal, isochoric, isobaric and isochoric, respectively. The total work done during cycle I is W_I and that during cycle II is W_{II}. The ratio W_I / W_{II} is _____. **[JEE (Advanced) 2023]**



PEH312

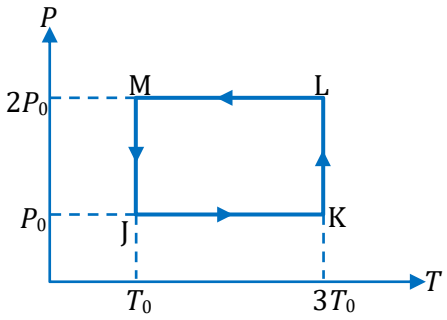
32. The specific heat capacity of a substance is temperature dependent and is given by the formula $C = kT$, where k is a constant of suitable dimensions in SI units, and T is the absolute temperature. If the heat required to raise the temperature of 1 kg of the substance from $-73°C$ to $27°C$ is nk, the value of n is _____. [Given : 0 K = −273 °C.]

[JEE (Advanced) 2024]

PEH220

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33. One mole of a monatomic ideal gas undergoes the cyclic process J → K → L → M → J, as shown in the P-T diagram.



Match the quantities mentioned in List-I with their values in List-II and choose the correct option. [R is the gas constant.]

	List-I		**List-II**
(P)	Work done in the complete cyclic process	(1)	$RT_0 - 4RT_0 \ln 2$
(Q)	Change in the internal energy of the gas in the process JK	(2)	0
(R)	Heat given to the gas in the process KL	(3)	$3RT_0$
(S)	Change in the internal energy of the gas in the process MJ	(4)	$-2RT_0 \ln 2$
		(5)	$-3RT_0 \ln 2$

[JEE (Advanced) 2024]

(A) P → 1; Q → 3; R → 5; S → 4
(B) P → 4; Q → 3; R → 5; S → 2
(C) P → 4; Q → 1; R → 2; S → 2
(D) P → 2; Q → 5; R → 3; S → 4

PEH221

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	D	B	D	A	A	B	D	A	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	D	C	C	C	A	B	C	A	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	A	B	A	C	C	D	B	B	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	B	B	C	D	A	A	B	C	C	A
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	B	B	A	B	C	D	D	B	C	C
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	B	C	C	C	D	B	A	A	D	D
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	C	C	D	D	C	A	B	B	C	A
Que.	71	72	73	74	75	76	77	78	79	80
Ans.	B	A	B	A	B	D	A	A	A	A
Que.	81	82	83							
Ans.	A	A	A							

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EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	AB	ABD	BD	BC	ACD or ABCD	BCD	AD	AC	CD	ACD
Que.	11	12	13	14	15	16	17	18	19	
Ans.	BC	ABC	AB	ABCD	B	A	B	D	D	
Que.	20					21				
Ans.	(A-P,S,T), (B-P,S,T), (C-Q,R), (D-Q,S)					(A-R), (B-P), (C-S), (D-Q)				
Que.	22									
Ans.	(A-P), (B-Q), (C-Q)									

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EXERCISE (S)

1. (a) $\theta = \frac{\pi}{2}$ (b) $\theta = \pi/4$. **2.** Strain in wire $A = 10^{-4}$ and strain in wire $B = 2 \times 10^{-4}$

3. (i) 1.67, (ii) 2.4 **4.** $\Delta L = \frac{3MgL}{2AY}$

5. (i) 50 N, (ii) 0.045 J, 1.8×10^{-3} m (iii) 8.4×10^{-4} m, (iv) $x = 0.12$ m

6. 1 : 1.26 **7.** 5 sec slow **8.** 5.8×10^{8} Pa

9. 4×10^{-6} $m/°C$ **10.** $F = \frac{(\alpha_1 L_1 + \alpha_2 L_2)t}{\left(\frac{L_1}{Y_1 A_1} + \frac{L_2}{Y_1 A_2}\right)}$ **11.** 2.0120 m

12. 7/2 **13.** (a) – 100°C (b) 1000 J **14.** $4/3\omega$

15. 9 **16.** 10.34 cm **17.** $\frac{K\pi r_1 r_2 (\theta_2 - \theta_1)}{L}$

18. $(T_2 - T_1)e^{-\frac{KA}{L}\left(\frac{m_1 s_1 + m_2 s_2}{m_1 s_1 m_2 s_2}\right)t}$ **19.** $\ln_2 \frac{Lms}{2KA}$ **20.** $3w/m - °C$

21. 166.3 sec **22.** 2 : 1 **23.** 9

24. 1000 J $(C°)^{-1}$ **25.** 887J **26.** $\left(\Delta T = \frac{mv_0^2}{3R}\right)$

27. 327°C **28.** 40 **29.** 32°C

30. $\left(\ell = \frac{21840}{673} cm\right)$ **31.** $P = \frac{10}{3} \times 10^5 N/m^2$, $T = \frac{10500}{31}$ $K \simeq 338.71$ K

32. (a) $T_{max} = \frac{2}{3}(p_0/R)\sqrt{p_0/3\alpha}$] (b) $T_{max} = p_0/e\beta R$

33. $m = (1 - e^{-Mgh/RT})\, p_0 S/g$ **34.** 12.7 kg

35. (a) 600 K, 900 K; (b) 10 J; (c) 14.9 J, 24.9 J; (d) 29.8 J

36. 1.5 **37.** $Q_1 < Q_2$

38. (a) $2T_0$, $\frac{p_0}{2}$ in the vessel A and $\frac{T_0}{2^{\gamma-2}}$, $p_0/2^{\gamma}$ in vessel B, (b) $2T_0$, $P_0/2$

39. (a) zero, (b) $\frac{3P_0}{2}$ on the left and $\frac{3P_0}{2}$ on the right, (c) $\frac{3T_0}{2}$, (d) $\frac{3P_0V_0}{4}$,

40. (a) $\frac{2p_1^{1/\gamma}V_0}{A}$, $\frac{2p_2^{1/\gamma}V_0}{A}$, (b) zero, (c) $(A/2)^{\gamma}$ where $A = p_1^{1/\gamma} + p_2^{1/\gamma}$

41. $\frac{nR}{P_a A}(T_S - T_0) + \left(1 - e^{-\frac{2KAt}{5nRx}}\right)$

42. (a) $P_i = P_a$, (b) $P_f = P_a + k(V - V_0)/A^2$ $V_0 = \ell A$, (c) $\Delta Q = P_a(V - V_0) + \frac{1}{2}k(V - v_0)^2 + C_v(T - T_0)$ $A = 1$

43. 1.6 m, 364 K **44.** 2 **45.** 2400

46. 5.7 J **47.** (a) 19 J/K.mol); (b) 25 J/(K.mol)

48. Approx. 4 J/K

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	1	2	4	2	4	20	1	3	57
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	40	1	5	3	4	2	1	40	4

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	AB	B	9	A	C	C	ABD	B	A
Que.	11	12	13	14	15	16	17	18	19	
Ans.	D	4	BCD	C	2.00	270.00	ACD	D	1.77 to 1.78	
Que.	20	21	22	23	24	25	26	27	28	29
Ans.	BCD	8.33	9	A	B	B	BCD	A	A	121
Que.	30	31	32	33						
Ans.	C	2	25000	B						

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